

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7402

Nombre de lignes df clean	7402
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5181
Taille dataset test	2220
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_1h_mean, r_NH3_st_1h_std, r_NH3_st_1h_rms, r_NH3_st_1h_median, r_NH3_st_1h_min, r_NH3_st_1h_max, r_NH3_st_1h_diff_last, r_NH3_st_1h_slope, r_temperature_st_1h_mean, r_temperature_st_1h_std, r_temperature_st_1h_rms, r_temperature_st_1h_median, r_temperature_st_1h_min, r_temperature_st_1h_max, r_temperature_st_1h_diff_last, r_temperature_st_1h_slope, r_umidity_st_1h_mean, r_umidity_st_1h_std, r_umidity_st_1h_rms, r_umidity_st_1h_median, r_umidity_st_1h_min, r_umidity_st_1h_max, r_umidity_st_1h_diff_last, r_umidity_st_1h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

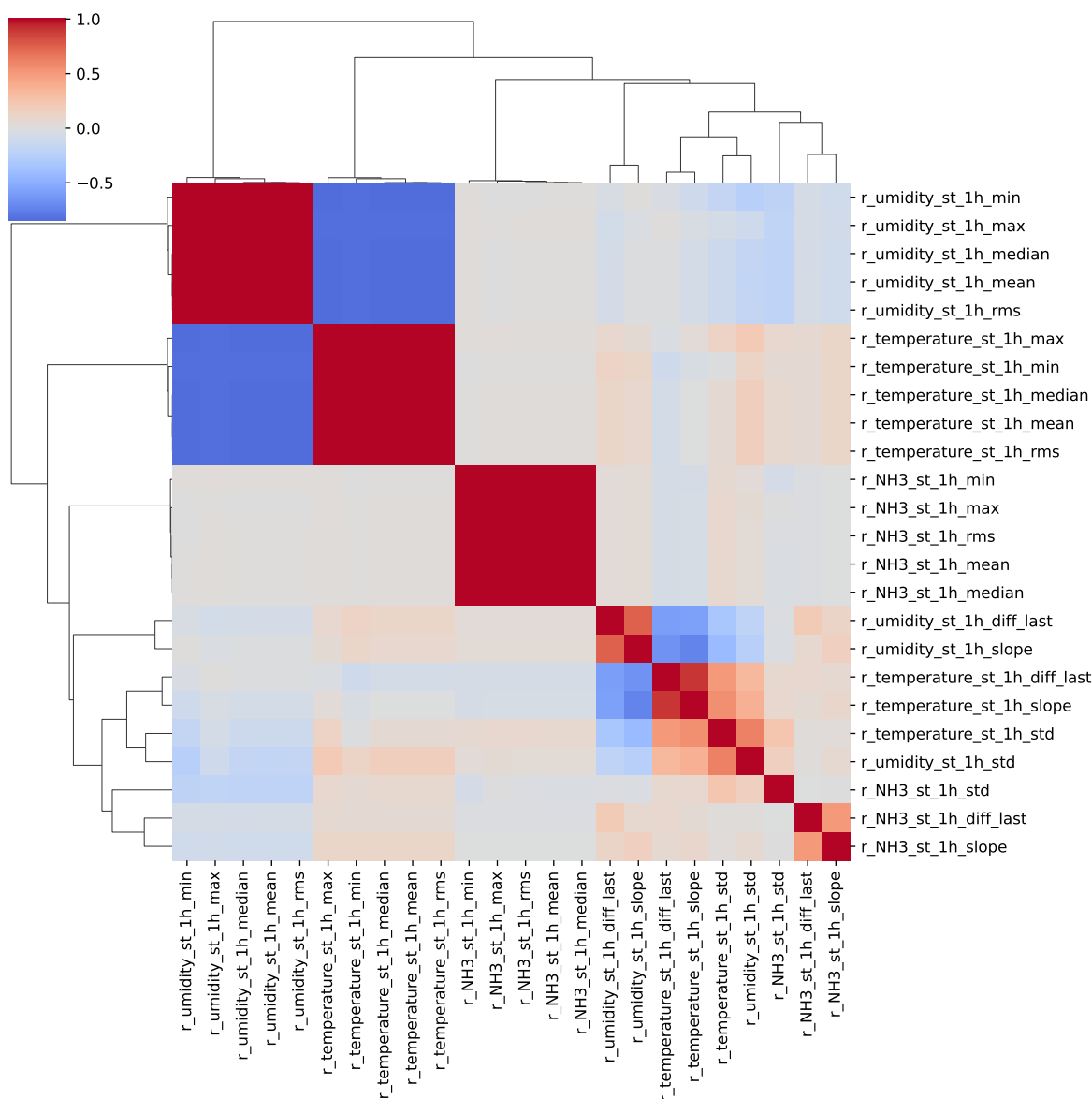
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_1h_mean, r_NH3_st_1h_max, r_NH3_st_1h_min, r_NH3_st_1h_rms, r_NH3_st_1h_median ; **cluster 7:** r_NH3_st_1h_std ; **cluster 8:** r_NH3_st_1h_diff_last ; **cluster 9:** r_NH3_st_1h_slope ; **cluster 10:** r_temperature_st_1h_min, r_umidity_st_1h_mean, r_temperature_st_1h_rms, r_umidity_st_1h_min, r_temperature_st_1h_max, r_temperature_st_1h_mean, r_umidity_st_1h_median, r_umidity_st_1h_rms, r_temperature_st_1h_median, r_umidity_st_1h_max ; **cluster 11:** r_temperature_st_1h_std ; **cluster 12:** r_temperature_st_1h_diff_last, r_temperature_st_1h_slope ; **cluster**

13: r_umidity_st_1h_std ; **cluster 14:** r_umidity_st_1h_diff_last ; **cluster 15:** r_umidity_st_1h_slope ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_1h_mean, r_NH3_st_1h_std, r_NH3_st_1h_diff_last, r_NH3_st_1h_slope, r_temperature_st_1h_min, r_temperature_st_1h_std, r_temperature_st_1h_diff_last, r_umidity_st_1h_std, r_umidity_st_1h_diff_last, r_umidity_st_1h_slope



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_1h_mean, r_NH3_st_1h_std, r_NH3_st_1h_diff_last, r_NH3_st_1h_slope, r_tempera-

ture_st_1h_min, r_temperature_st_1h_std, r_temperature_st_1h_diff_last, r_umidity_st_1h_std, r_umidity_st_1h_diff_last, r_umidity_st_1h_slope

Len(Features_X): 15

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

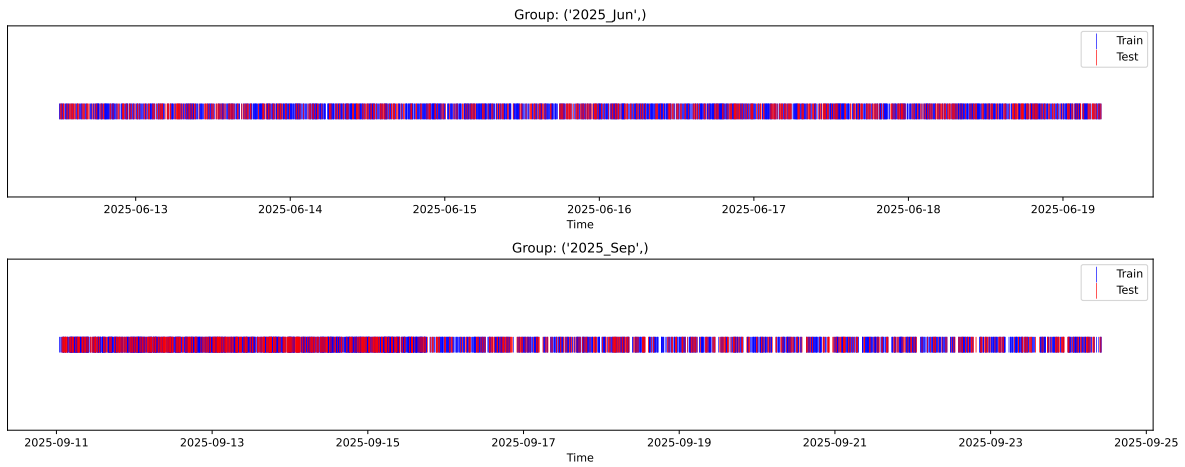
External Split: Train: 5181 samples Test: 2221 samples



3.3.2 Internal CV Folds

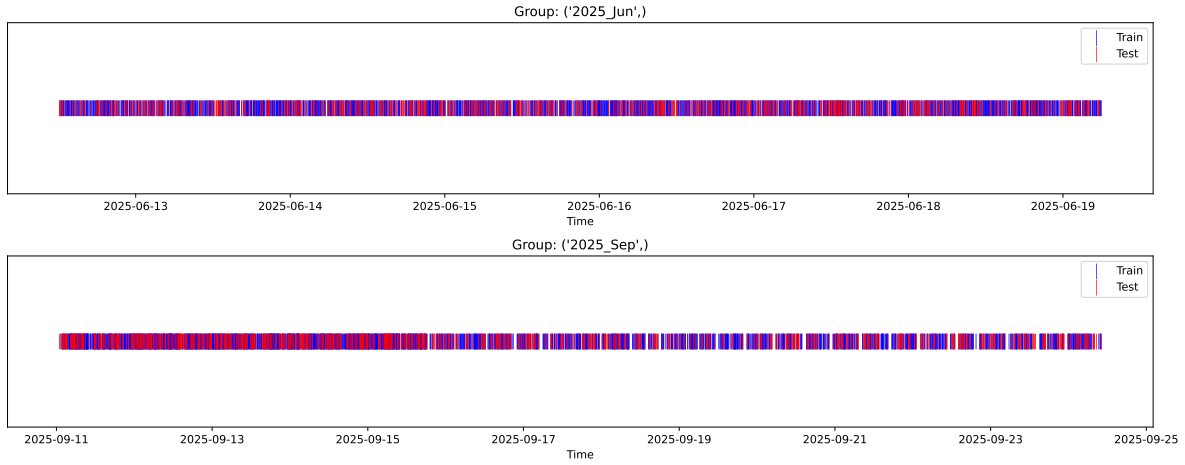
Fold 1/5

Train: 3626 samples Test: 1555 samples Excluded: 0 samples Window exclusion: 0h



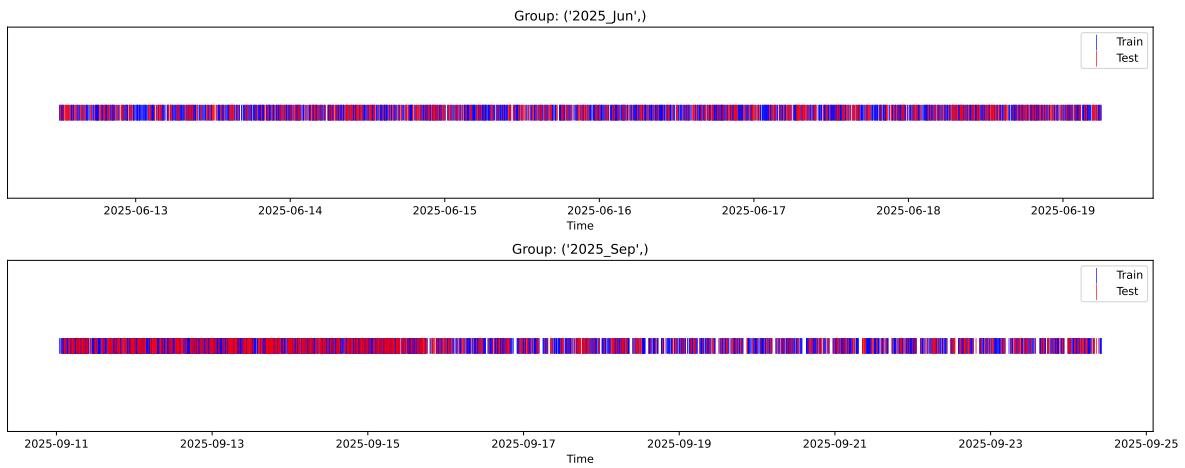
Fold 2/5

Train: 3626 samples Test: 1555 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3626 samples Test: 1555 samples Excluded: 0 samples Window exclusion: 0h



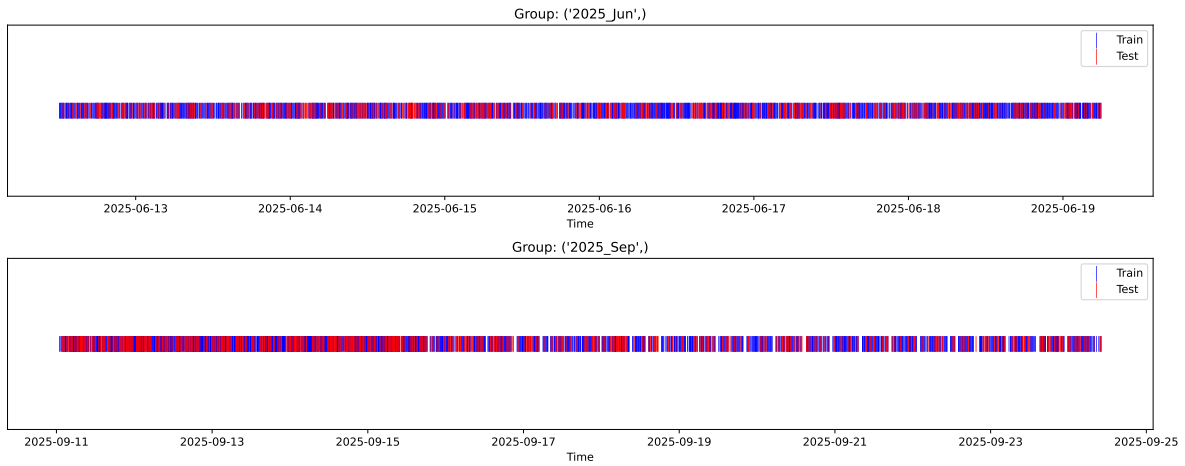
Fold 4/5

Train: 3626 samples Test: 1555 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3626 samples Test: 1555 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.358, std=0.783, min=0.959, max=8.190
 y_{test} : mean=2.367, std=0.829, min=1.041, max=13.845

Fold 0 — y_{test} : mean=2.33, std=0.76, y_{train} : mean=2.37, std=0.79

Fold 1 — y_{test} : mean=2.36, std=0.79, y_{train} : mean=2.36, std=0.78

Fold 2 — y_{test} : mean=2.37, std=0.79, y_{train} : mean=2.35, std=0.78

Fold 3 — y_test: mean=2.37, std=0.80, y_train: mean=2.35, std=0.77

Fold 4 — y_test: mean=2.34, std=0.77, y_train: mean=2.36, std=0.79

train scores CV: [0.52518328 0.53758093 0.53284027 0.54169525 0.52887757]

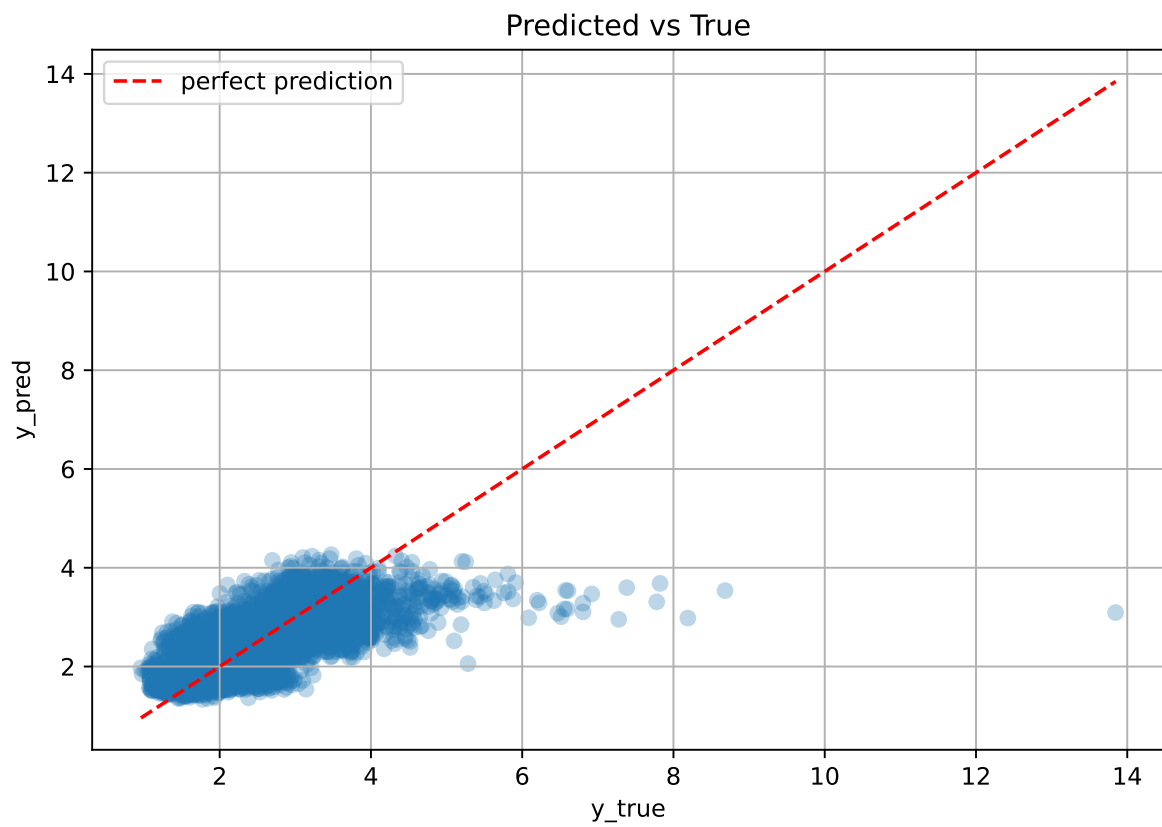
test scores CV: [0.54750977 0.51944822 0.52959449 0.50951963 0.5390501]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5332	0.529	0.0135	0.5089	0.5094	-0.0196	0.9629	1.008

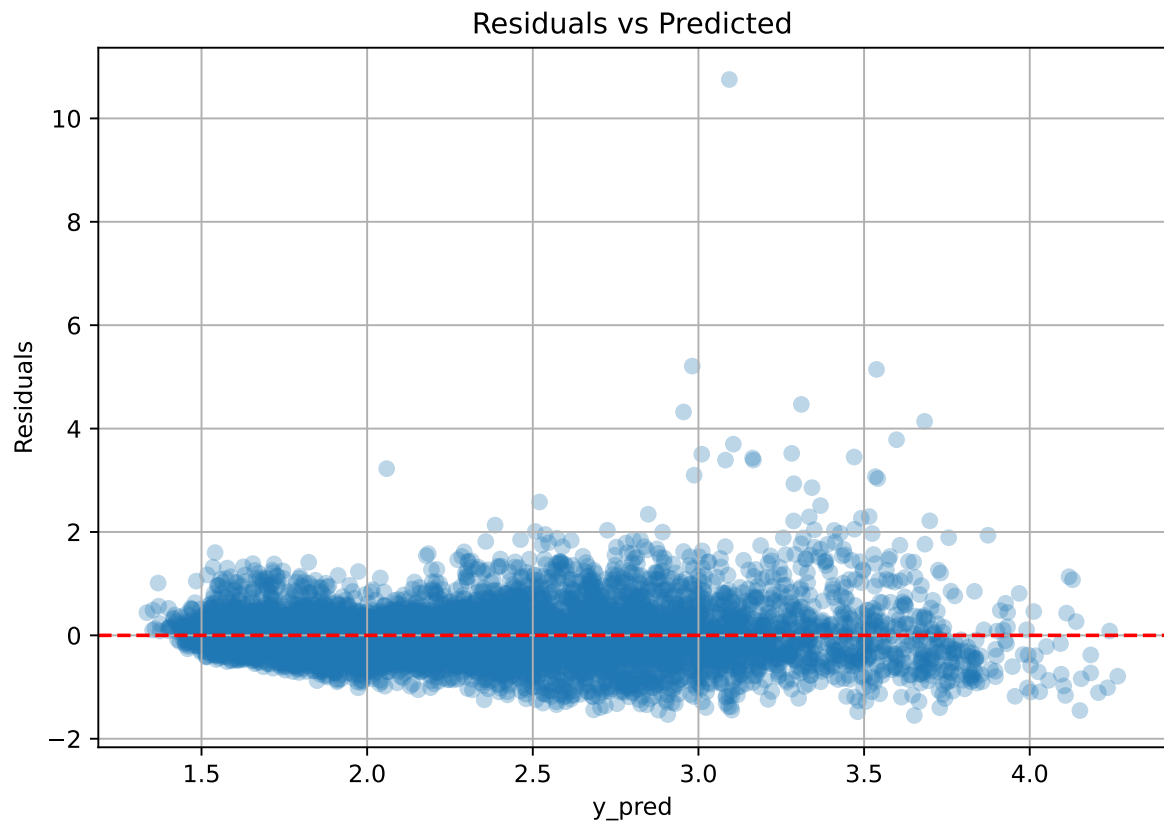
Coefficients:

coef_r_CO2 = -0.0498, coef_r_NH3 = 0.1067, coef_r_NH3_st_1h_diff_last = 0.0164, coef_r_NH3_st_1h_mean = -0.1429, coef_r_NH3_st_1h_slope = 0.0109, coef_r_NH3_st_1h_std = 0.0699, coef_r_THI = 0.3666, coef_r_temperature = 1.1426, coef_r_temperature_st_1h_diff_last = -0.0724, coef_r_temperature_st_1h_min = -1.5167, coef_r_temperature_st_1h_std = -0.0347, coef_r_umidity = -0.4330, coef_r_umidity_st_1h_diff_last = -0.0175, coef_r_umidity_st_1h_slope = -0.0593, coef_r_umidity_st_1h_std = 0.0649, intercept = 2.3579,

3.4.2 Scatter



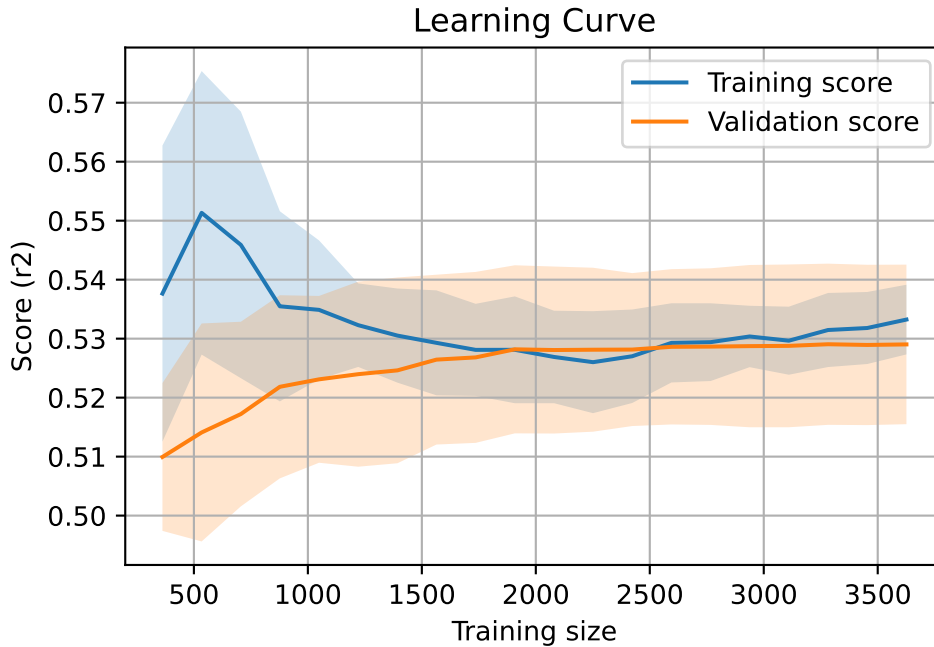
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7402, **Len(training)** : 5182,

Len(training_real) : 3628, **Len(test_of_training)** : 1554, **Len(test)** : 2220

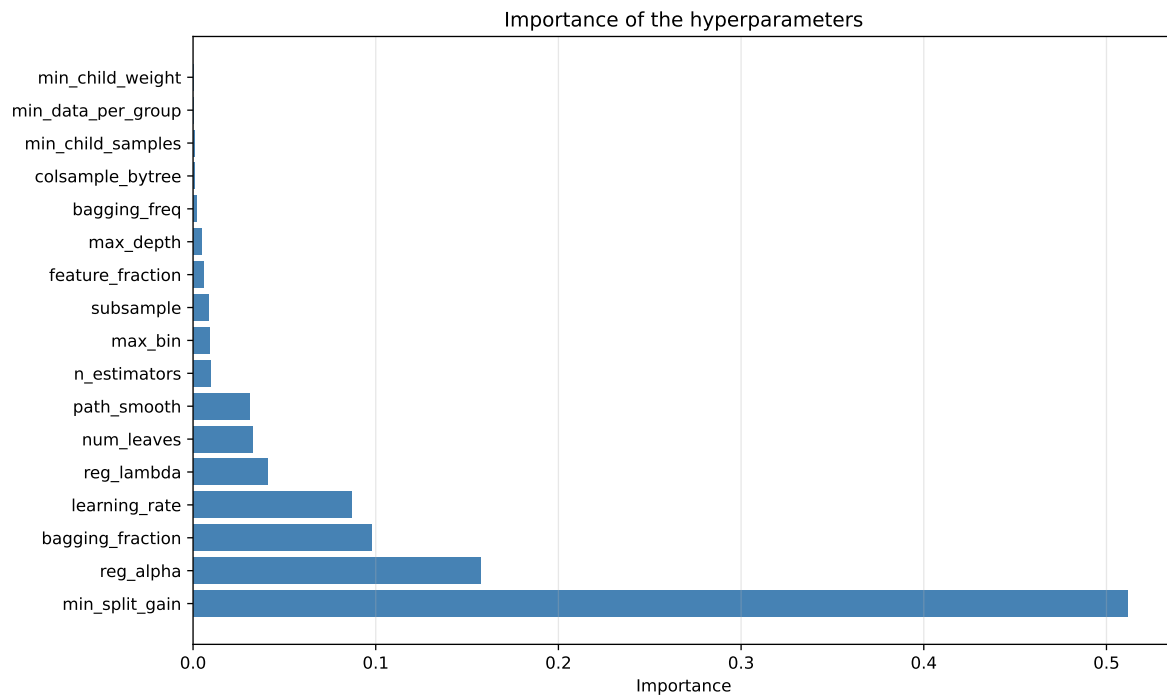
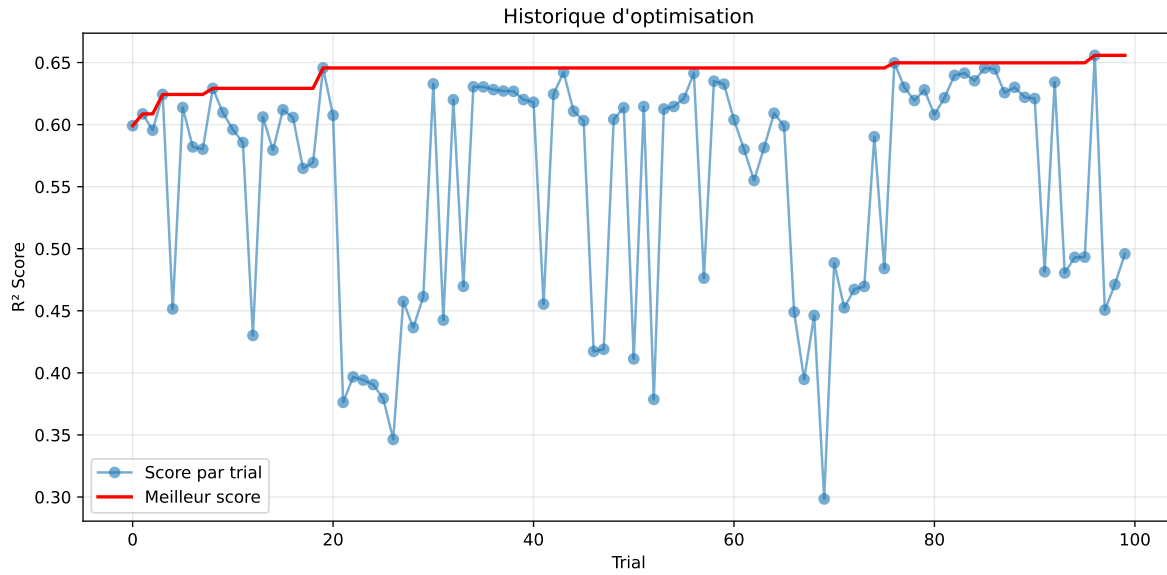


3.5 Model : LGBM

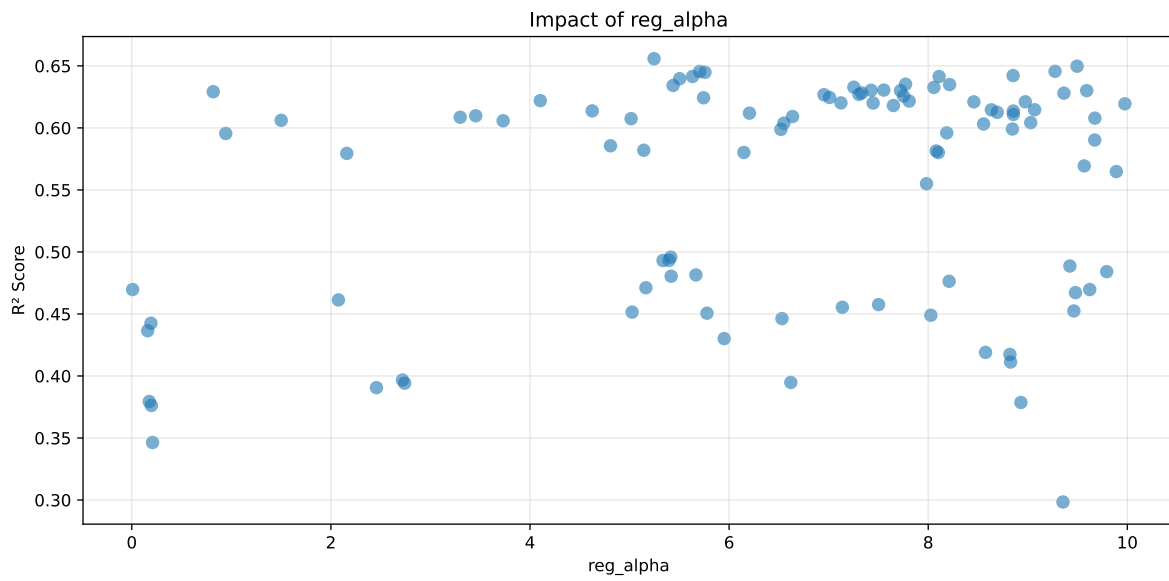
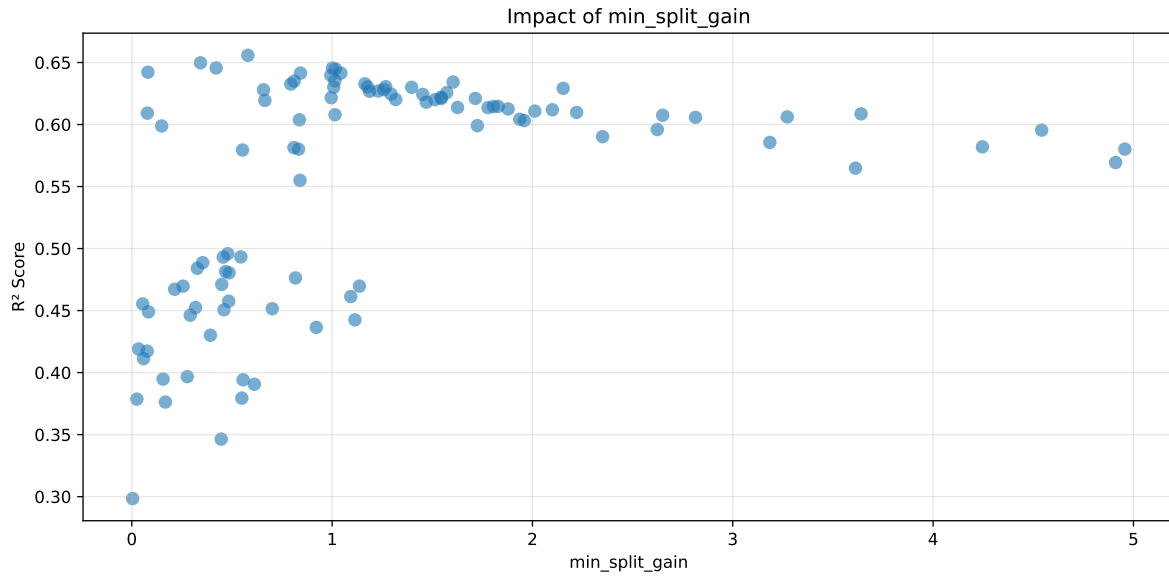
3.5.1 Gridsearch

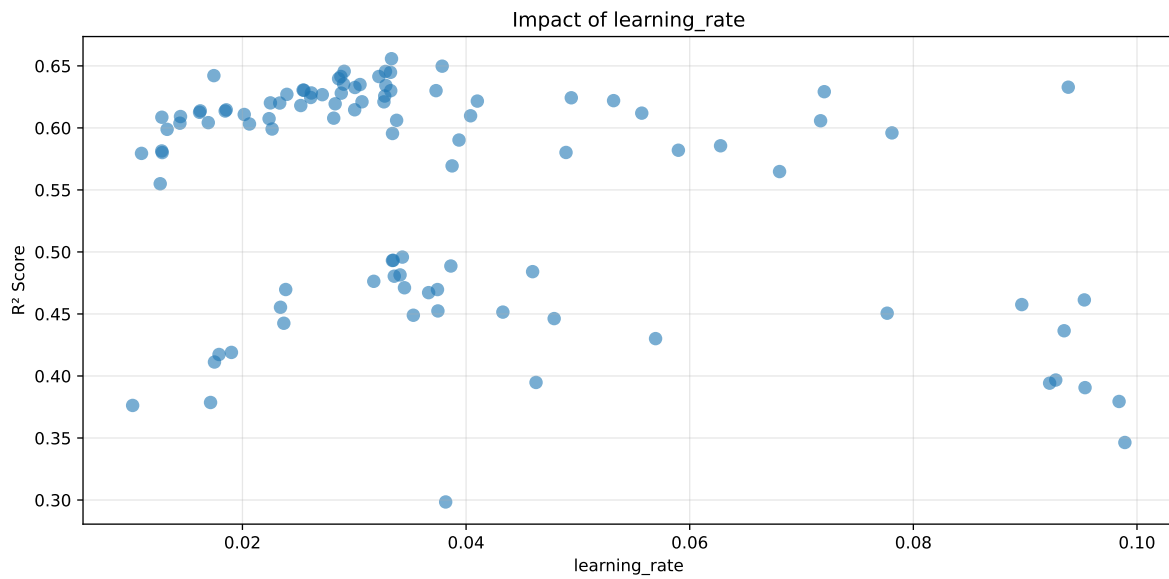
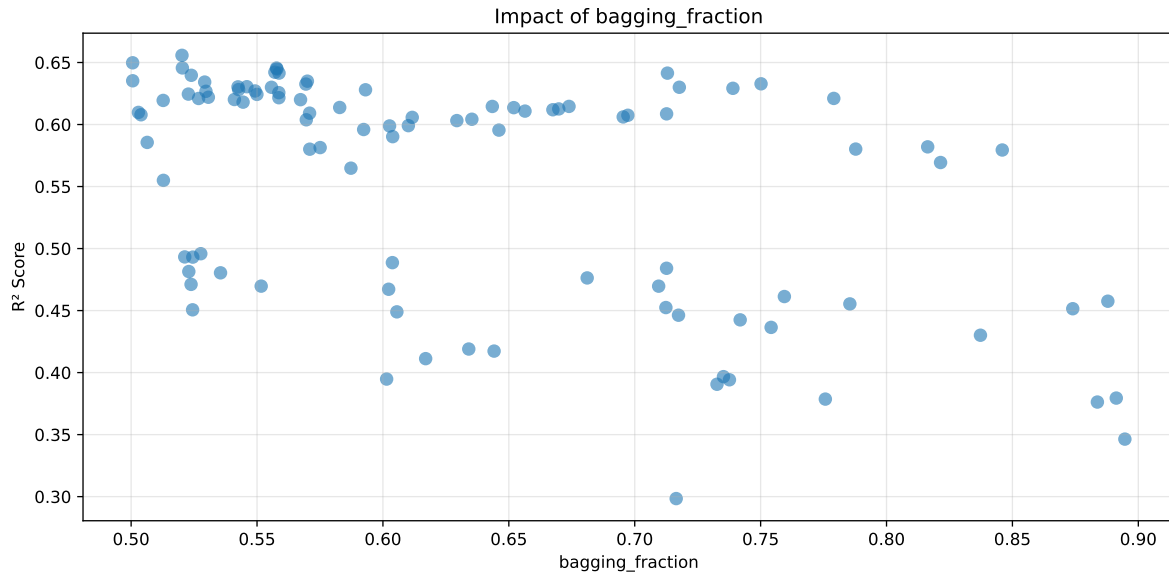
Distribution y_train vs y_test: y_train: mean=2.358, std=0.783, min=0.959, max=8.190
y_test: mean=2.367, std=0.829, min=1.041, max=13.845

===== TOP Importance FEATURES =====
feature importance r_CO2 460 r_NH3 282 r_umidity 220 r_temperature 200 r_NH3_st_1h_mean
165 r_NH3_st_1h_std 154 r_temperature_st_1h_diff_last 150 r_temperature_st_1h_std
143 r_umidity_st_1h_slope 115 r_temperature_st_1h_min 103 r_THI 101 r_umid-
ity_st_1h_std 83 r_NH3_st_1h_diff_last 70 r_NH3_st_1h_slope 52 r_umid-
ity_st_1h_diff_last 47



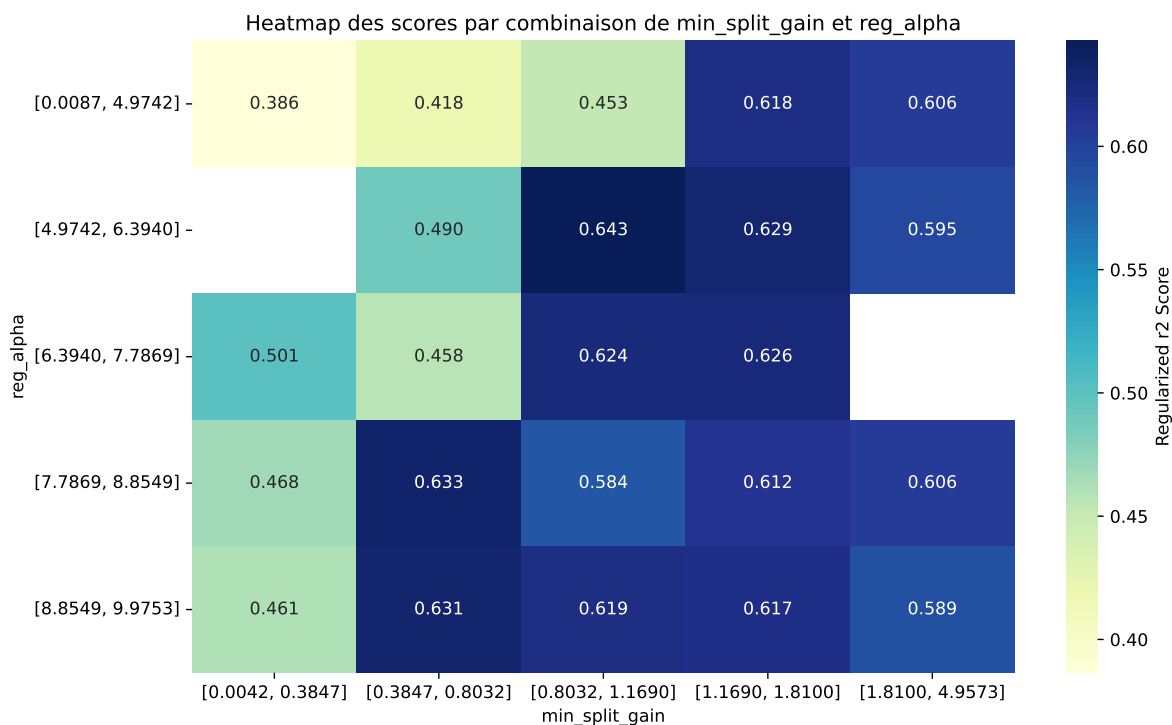
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'bagging_fraction', 'learning_rate']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6855	0.6558	0.6117	0.0147	0.6558	0.6349	-0.0209	0.9681	1.0452
2	0.6788	0.6498	0.6047	0.0132	0.6498	0.6258	-0.024	0.9631	1.0447
3	0.6717	0.6456	0.6003	0.0104	0.6456	0.6236	-0.022	0.9659	1.0403
4	0.6746	0.6455	0.6042	0.0137	0.6455	0.627	-0.0185	0.9713	1.0451
5	0.6738	0.6448	0.6038	0.0134	0.6448	0.6248	-0.02	0.969	1.045
6	0.6719	0.6422	0.5989	0.0131	0.6422	0.615	-0.0271	0.9577	1.0464
7	0.6683	0.6415	0.6	0.0135	0.6415	0.6211	-0.0204	0.9682	1.0418
8	0.6717	0.6414	0.5989	0.0104	0.6414	0.616	-0.0254	0.9604	1.0473
9	0.6668	0.6397	0.5992	0.0146	0.6397	0.6188	-0.0209	0.9673	1.0424
10	0.6573	0.6353	0.5927	0.0144	0.6353	0.614	-0.0213	0.9664	1.0347
---	---	---	---	---	---	---	---	---	---
100	0.7871	0.7057	0.6609	0.0138	0.2984	0.6836	-0.0221	0.9687	1.1154

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.033330536801801806, 'reg__num_leaves': 39, 'reg__subsample': 0.5228816737164611, 'reg__colsample_bytree': 0.6889971209058227, 'reg__min_child_samples': 41, 'reg__reg_alpha': 5.245948989073565, 'reg__reg_lambda': 29.87537717974598, 'reg__min_split_gain': 0.5793464141300757, 'reg__path_smooth': 4.278010527152889, 'reg__min_child_weight':

0.12113099078218421, 'reg__feature_fraction': 0.6683707789742226, 'reg__bagging_fraction': 0.5201966156312621, 'reg__bagging_freq': 1, 'reg__max_bin': 72, 'reg__min_data_per_group': 95}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 8, 'reg__learning_rate': 0.037863465717651756, 'reg__num_leaves': 31, 'reg__subsample': 0.640258679451962, 'reg__colsample_bytree': 0.6739797357380002, 'reg__min_child_samples': 31, 'reg__reg_alpha': 9.49509294313733, 'reg__reg_lambda': 33.58687284031856, 'reg__min_split_gain': 0.34288317602504265, 'reg__path_smooth': 4.772828477487355, 'reg__min_child_weight': 0.19044597616272912, 'reg__feature_fraction': 0.6878134420276785, 'reg__bagging_fraction': 0.5005967771833407, 'reg__bagging_freq': 2, 'reg__max_bin': 126, 'reg__min_data_per_group': 72}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.029098624436482663, 'reg__num_leaves': 63, 'reg__subsample': 0.6498937221182702, 'reg__colsample_bytree': 0.6784366538954272, 'reg__min_child_samples': 21, 'reg__reg_alpha': 9.274012432178386, 'reg__reg_lambda': 19.870198225102897, 'reg__min_split_gain': 0.4211262874342503, 'reg__path_smooth': 4.725820474919881, 'reg__min_child_weight': 6.600185357664305, 'reg__feature_fraction': 0.7777850016287893, 'reg__bagging_fraction': 0.5203510557282638, 'reg__bagging_freq': 3, 'reg__max_bin': 71, 'reg__min_data_per_group': 73}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.03278960845757779, 'reg__num_leaves': 36, 'reg__subsample': 0.5533990681567652, 'reg__colsample_bytree': 0.6725331498374292, 'reg__min_child_samples': 13, 'reg__reg_alpha': 5.7043382035999315, 'reg__reg_lambda': 25.050089682873715, 'reg__min_split_gain': 1.0017108377488277, 'reg__path_smooth': 4.307083596417254, 'reg__min_child_weight': 0.1765280941362089, 'reg__feature_fraction': 0.8112320659055952, 'reg__bagging_fraction': 0.557740989418546, 'reg__bagging_freq': 1, 'reg__max_bin': 73, 'reg__min_data_per_group': 108}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.03324054656848526, 'reg__num_leaves': 18, 'reg__subsample': 0.5244610659126353, 'reg__colsample_bytree': 0.663763464745571, 'reg__min_child_samples': 13, 'reg__reg_alpha': 5.757860121616578, 'reg__reg_lambda': 25.719286241918724, 'reg__min_split_gain': 1.0158992340089388, 'reg__path_smooth': 4.3649315231369386, 'reg__min_child_weight': 0.4638307970099698, 'reg__feature_fraction': 0.8157462868721483, 'reg__bagging_fraction': 0.5579002807581608, 'reg__bagging_freq': 1, 'reg__max_bin': 73, 'reg__min_data_per_group': 108}

Rank 6: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.017445483697094463, 'reg__num_leaves': 21, 'reg__subsample': 0.6176058874430521, 'reg__colsample_bytree': 0.6397134132974581, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.852379602567574, 'reg__reg_lambda': 28.90155280788415, 'reg__min_split_gain': 0.08056230699190153, 'reg__path_smooth': 4.955620154729964, 'reg__min_child_weight':

0.045848513059659264, 'reg__feature_fraction': 0.8960871293412029, 'reg__bagging_fraction': 0.5570868482173604, 'reg__bagging_freq': 1, 'reg__max_bin': 84, 'reg__min_data_per_group': 52}

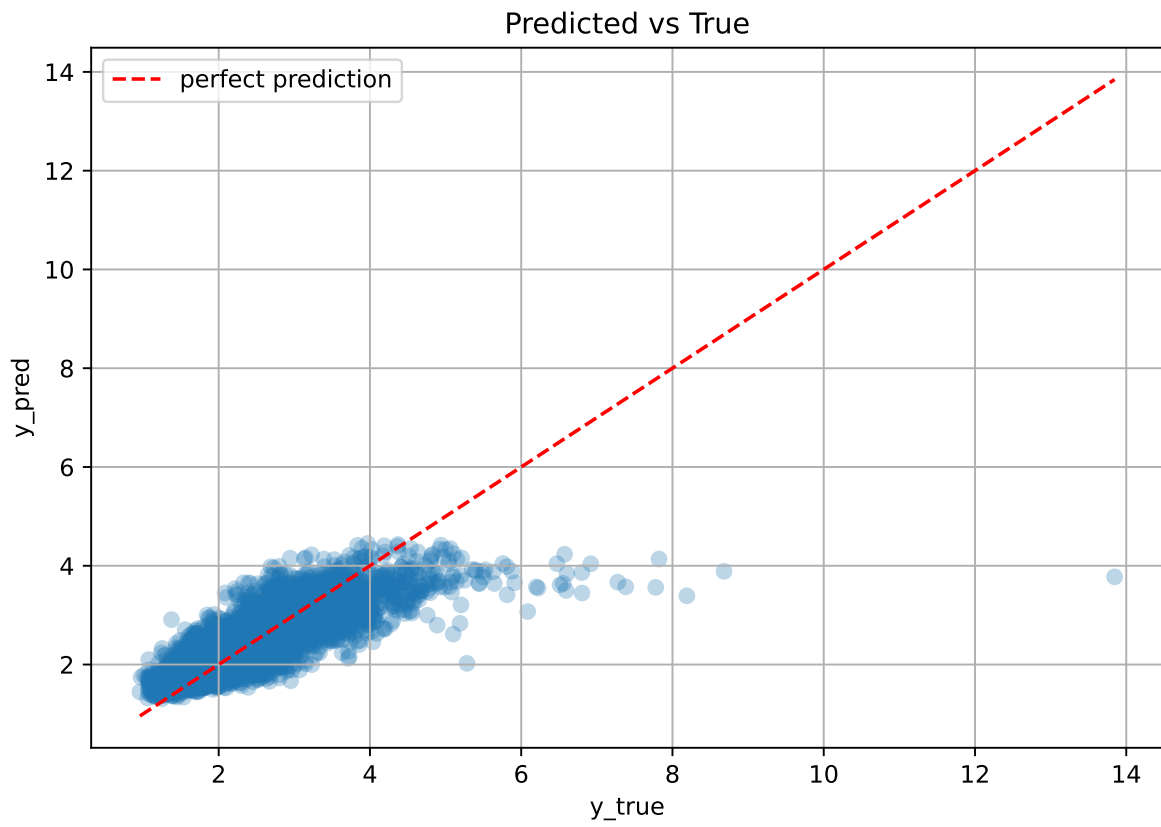
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.028799920126964817, 'reg__num_leaves': 22, 'reg__subsample': 0.6461342504610238, 'reg__colsample_bytree': 0.6811981281404468, 'reg__min_child_samples': 28, 'reg__reg_alpha': 5.632277695656765, 'reg__reg_lambda': 32.9332217535574, 'reg__min_split_gain': 1.0435760269591288, 'reg__path_smooth': 4.265223700541931, 'reg__min_child_weight': 0.04789191183728293, 'reg__feature_fraction': 0.8112465458048954, 'reg__bagging_fraction': 0.558731289507536, 'reg__bagging_freq': 1, 'reg__max_bin': 73, 'reg__min_data_per_group': 81}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.03219951248870245, 'reg__num_leaves': 25, 'reg__subsample': 0.6654812515961251, 'reg__colsample_bytree': 0.5757136006581238, 'reg__min_child_samples': 31, 'reg__reg_alpha': 8.10931612660762, 'reg__reg_lambda': 31.21595126966631, 'reg__min_split_gain': 0.8420308102534753, 'reg__path_smooth': 3.7520198934830993, 'reg__min_child_weight': 0.10277059421299485, 'reg__feature_fraction': 0.7440661142335706, 'reg__bagging_fraction': 0.7129922034399813, 'reg__bagging_freq': 2, 'reg__max_bin': 80, 'reg__min_data_per_group': 86}

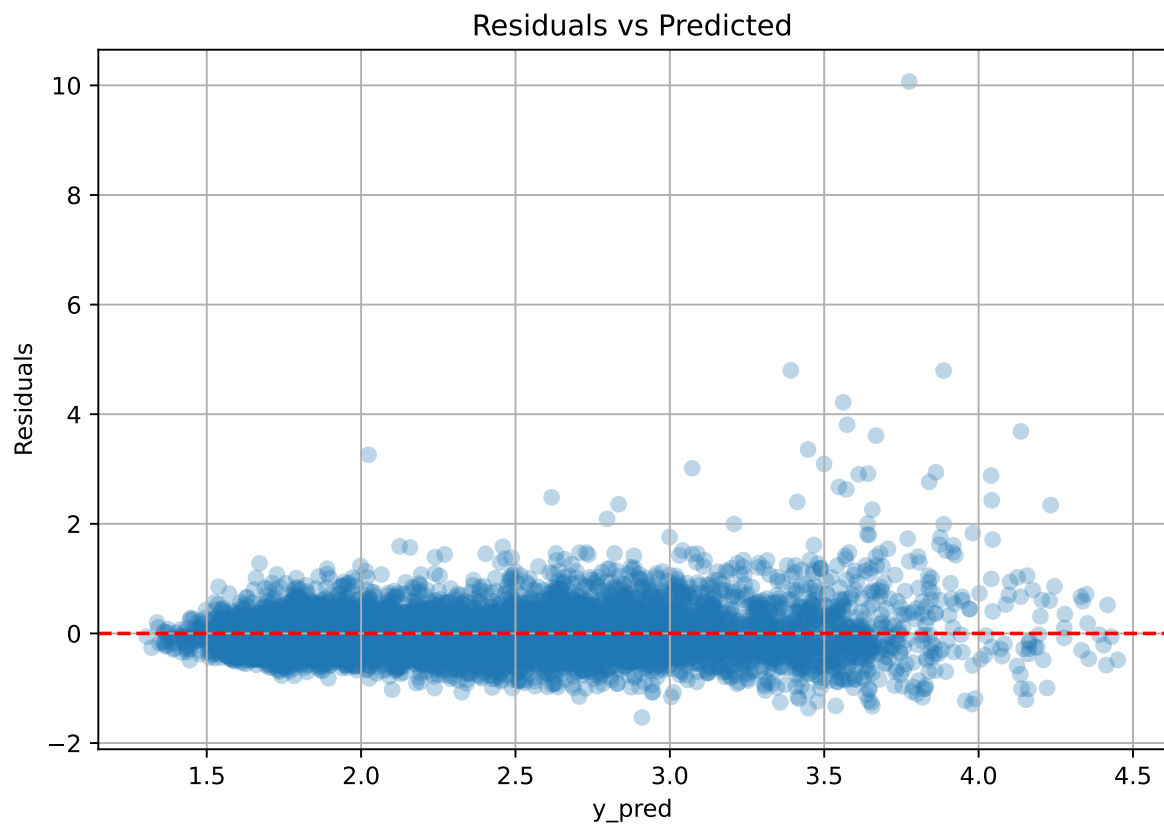
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.02859911529998689, 'reg__num_leaves': 22, 'reg__subsample': 0.528627935633621, 'reg__colsample_bytree': 0.6781130915805298, 'reg__min_child_samples': 36, 'reg__reg_alpha': 5.503493125073076, 'reg__reg_lambda': 33.12147260664599, 'reg__min_split_gain': 0.9935077204813876, 'reg__path_smooth': 4.761783327154191, 'reg__min_child_weight': 0.005823657142165671, 'reg__feature_fraction': 0.8786970819064969, 'reg__bagging_fraction': 0.5238979150060437, 'reg__bagging_freq': 1, 'reg__max_bin': 74, 'reg__min_data_per_group': 94}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.029053244703299622, 'reg__num_leaves': 22, 'reg__subsample': 0.5265925958231356, 'reg__colsample_bytree': 0.6982815335336358, 'reg__min_child_samples': 13, 'reg__reg_alpha': 7.772454837694768, 'reg__reg_lambda': 25.122489722356228, 'reg__min_split_gain': 1.0140291138730053, 'reg__path_smooth': 4.297377142469831, 'reg__min_child_weight': 0.16514700190075612, 'reg__feature_fraction': 0.816648853135105, 'reg__bagging_fraction': 0.500619712317841, 'reg__bagging_freq': 1, 'reg__max_bin': 71, 'reg__min_data_per_group': 79}

3.5.2 Scatter



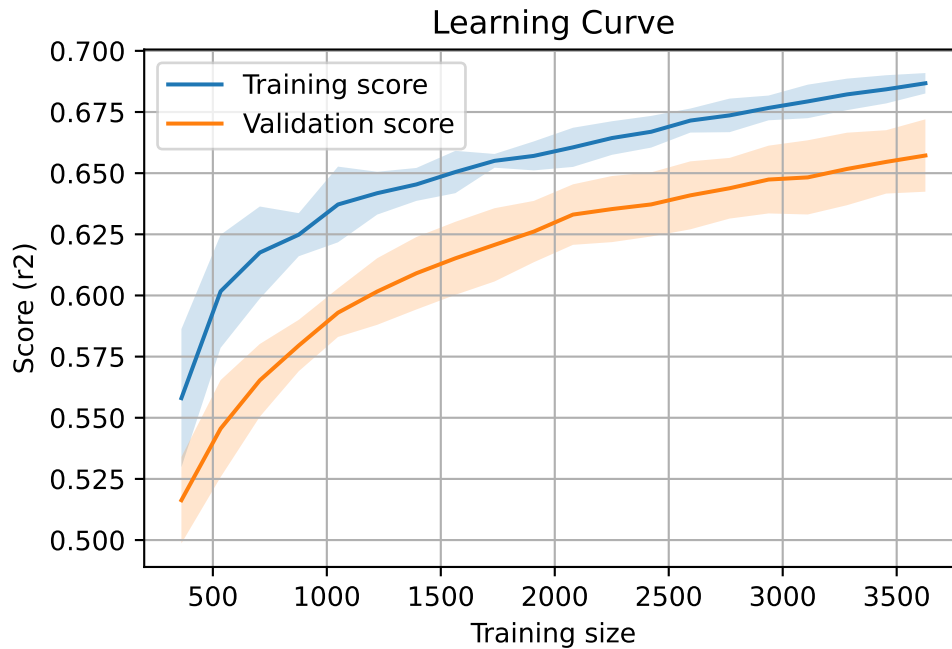
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7402, `Len(training)` : 5182,

`Len(training_real)` : 3628, `Len(test_of_training)` : 1554, `Len(test)` : 2220



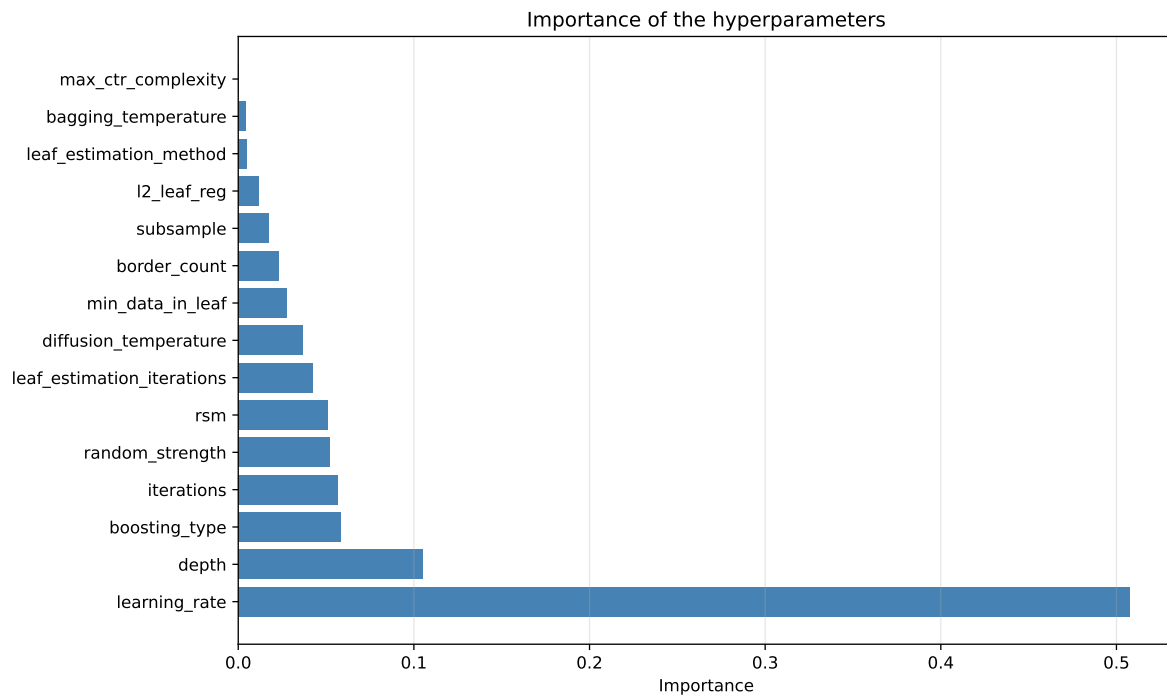
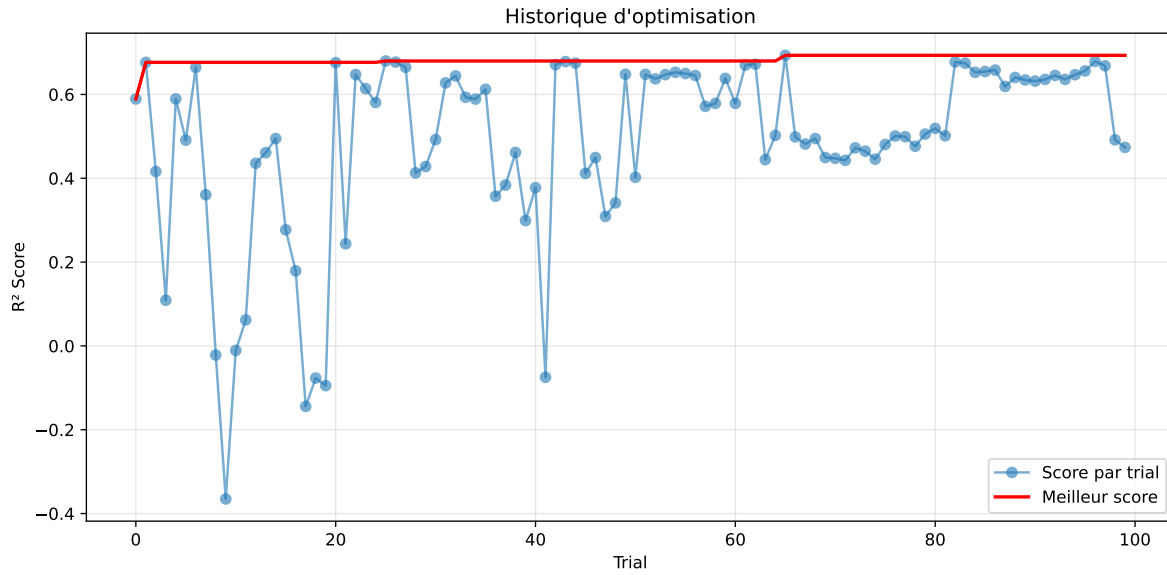
3.6 Model : CatBoost

3.6.1 Gridsearch

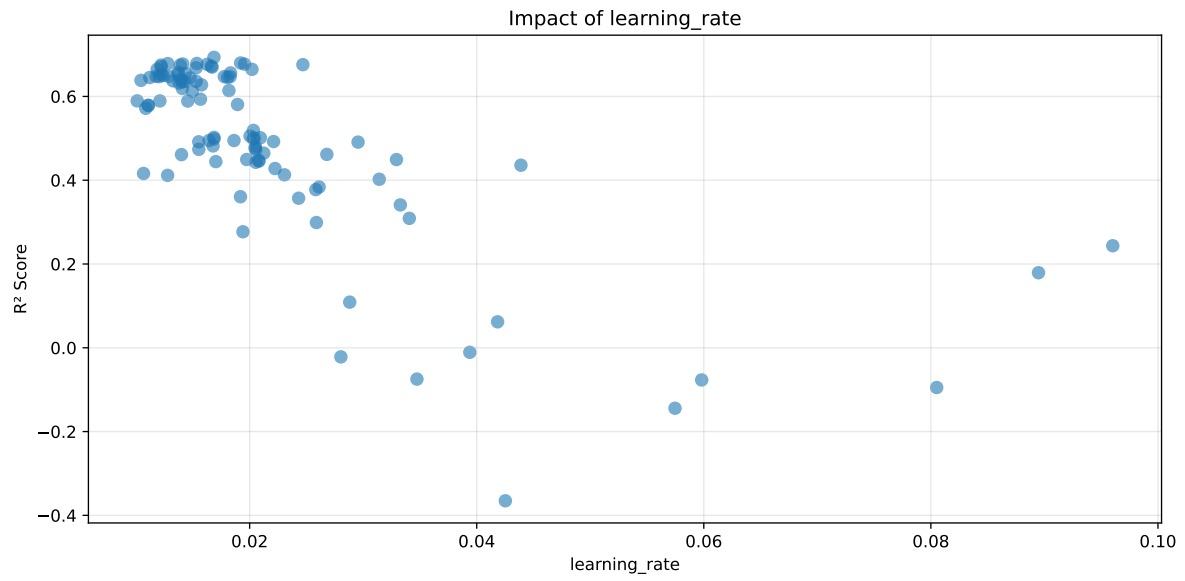
Distribution y_train vs y_test: y_train: mean=2.358, std=0.783, min=0.959, max=8.190
y_test: mean=2.367, std=0.829, min=1.041, max=13.845

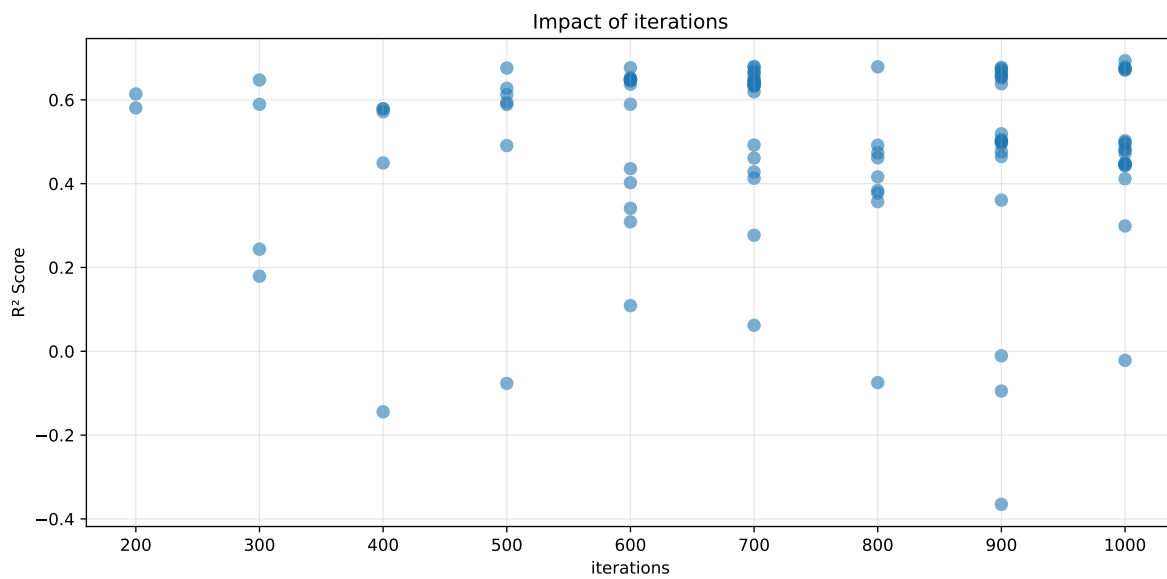
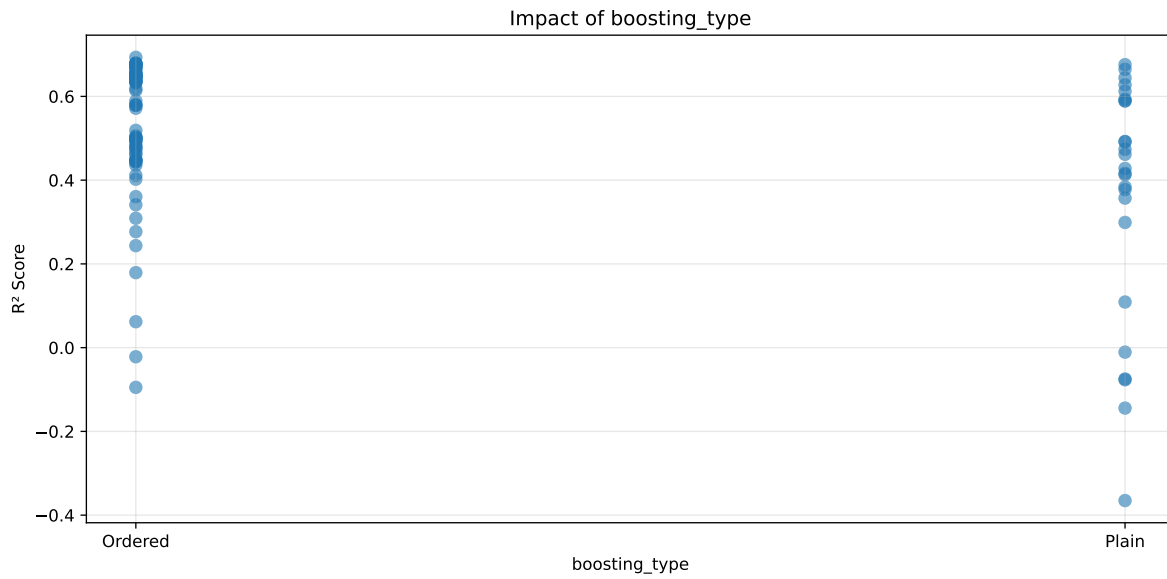
===== TOP Importance FEATURES =====

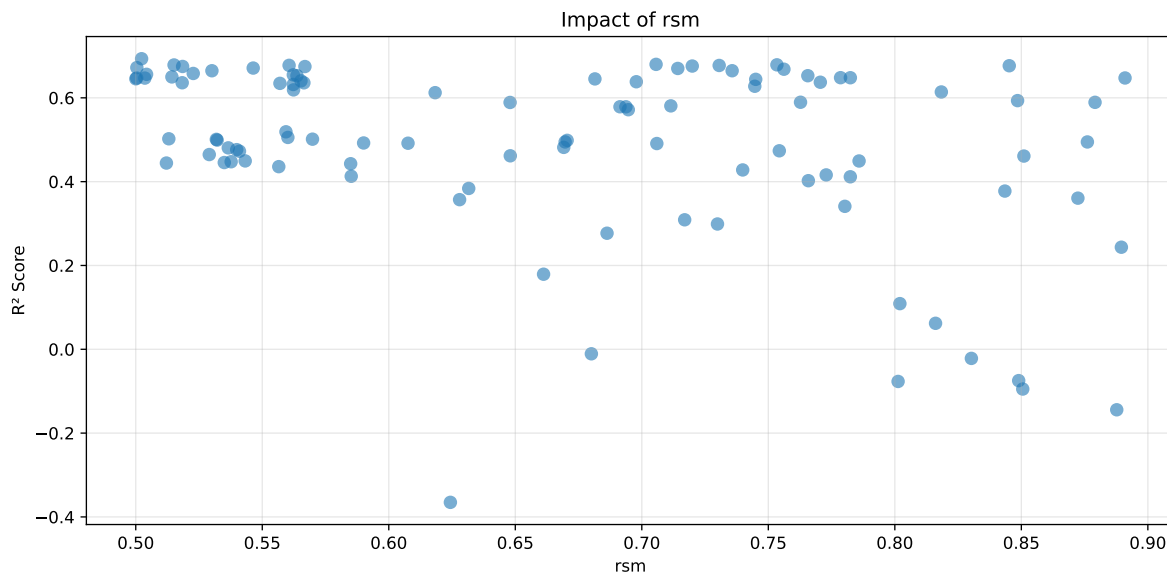
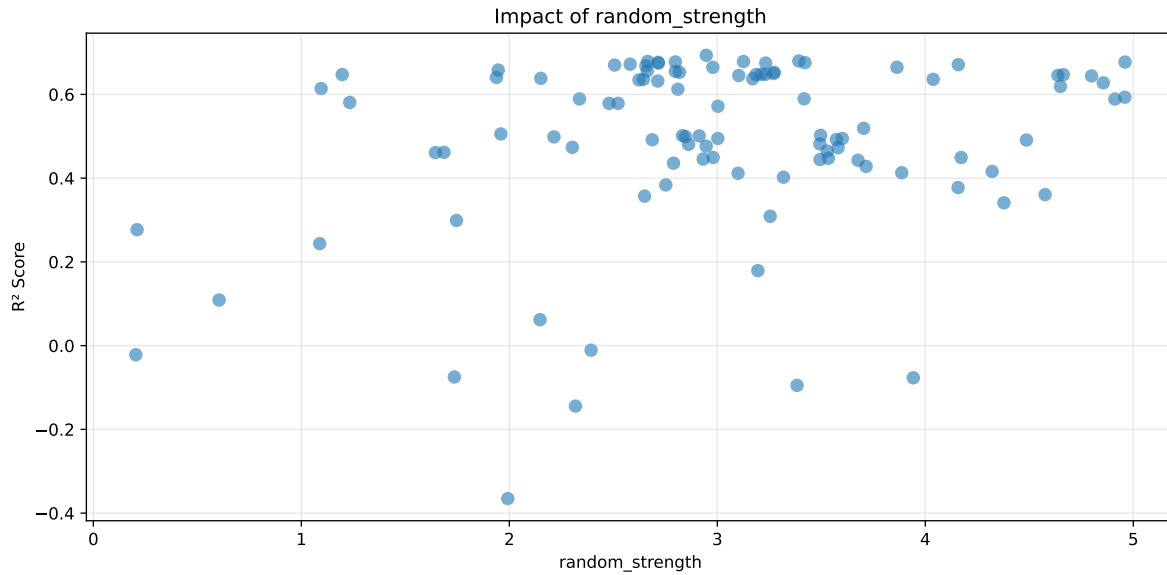
Feature	Importance
r_umidity	24.682626
r_CO2	11.595869
r_temperature_st_1h_diff_last	11.387412
r_NH3	7.959315
r_umidity_st_1h_slope	7.535837
r_temperature	7.208739
r_temperature_st_1h_std	6.293541
r_NH3_st_1h_mean	6.052187
r_THI	3.791747
r_NH3_st_1h_std	3.115645
r_temperature_st_1h_min	3.016834
r_umidity_st_1h_diff_last	2.092627
r_umidity_st_1h_std	1.964989
r_NH3_st_1h_slope	1.915992
r_NH3_st_1h_diff_last	1.386639



Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'boosting_type', 'iterations', 'random_strength', 'rsm']

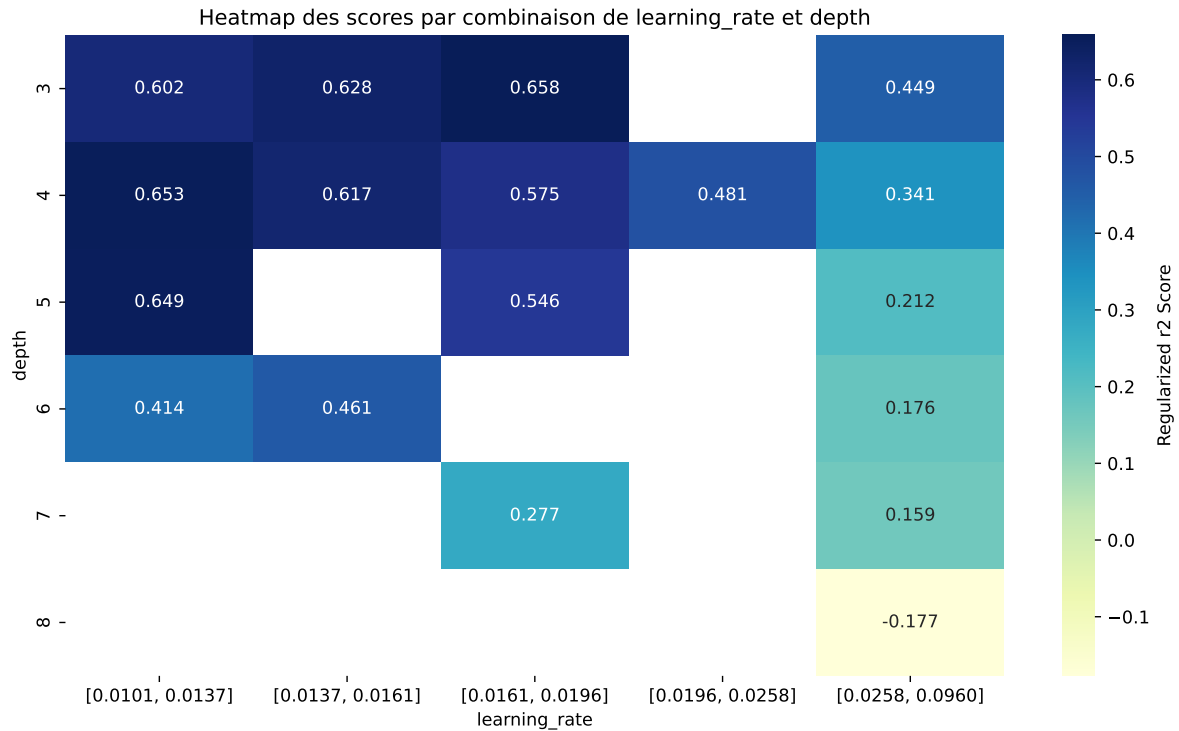






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7279	0.6933	0.644	0.0165	0.6933	0.6514	-0.0419	0.9395	1.0499
2	0.7113	0.6799	0.6331	0.0145	0.6799	0.6393	-0.0406	0.9403	1.0462
3	0.7027	0.6787	0.6282	0.0156	0.6787	0.6354	-0.0434	0.9361	1.0353
4	0.7062	0.6784	0.6312	0.0154	0.6784	0.637	-0.0414	0.939	1.041
5	0.7025	0.6775	0.6277	0.0156	0.6775	0.6335	-0.044	0.935	1.0368
6	0.705	0.6772	0.629	0.0136	0.6772	0.639	-0.0383	0.9435	1.041
7	0.7076	0.6766	0.6275	0.015	0.6766	0.632	-0.0446	0.9341	1.0458
8	0.7094	0.6759	0.6302	0.0136	0.6759	0.6368	-0.039	0.9423	1.0497
9	0.7006	0.6748	0.6254	0.0142	0.6748	0.63	-0.0449	0.9335	1.0382
10	0.6988	0.6747	0.6258	0.016	0.6747	0.6304	-0.0443	0.9343	1.0359
---	---	---	---	---	---	---	---	---	---
100	0.9716	0.7488	0.7103	0.0213	-0.3652	0.7254	-0.0234	0.9688	1.2976

Hyperparameters:

Rank 1: {'reg__iterations': 1000, 'reg__depth': 4, 'reg__learning_rate': 0.016856890308800693, 'reg__l2_leaf_reg': 32.78093380817312, 'reg__bagging_temperature': 0.9863985142833626, 'reg__random_strength': 2.946864740891236, 'reg__subsample': 0.6394674082483256, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.5023052134526383, 'reg__border_count': 36, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5468.914956695513}

Rank 2: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.019204109207098793, 'reg__l2_leaf_reg': 33.694356017319265, 'reg__bagging_temperature': 2.9316440356646973, 'reg__random_strength': 3.3920323777272294, 'reg__subsample': 0.8856616031764885, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7056408709874952, 'reg__border_count': 106, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2447.303568199954}

Rank 3: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.01535100958863014, 'reg__l2_leaf_reg': 31.520796478215257, 'reg__bagging_temperature': 3.170422930576377, 'reg__random_strength': 2.6651923966358044, 'reg__subsample': 0.7544662961180902, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7533983215473163, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3819.7836404299405}

Rank 4: {'reg__iterations': 1000, 'reg__depth': 4, 'reg__learning_rate': 0.012793282372370561, 'reg__l2_leaf_reg': 26.232435634299815, 'reg__bagging_temperature': 0.5996196562114466, 'reg__random_strength': 3.1257662638378347, 'reg__subsample': 0.7562789701218641, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5151316599187715, 'reg__border_count': 63, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4285.409562938092}

Rank 5: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.014106569255523714, 'reg__l2_leaf_reg': 25.09134443145831, 'reg__bagging_temperature': 3.7999381003519335, 'reg__random_strength': 2.798291133570864, 'reg__subsample': 0.6762003907111852, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.5605992558748109, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3878.274922121303}

Rank 6: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.019546974823459785, 'reg__l2_leaf_reg': 33.31601497989705, 'reg__bagging_temperature': 3.1857602861524548, 'reg__random_strength': 4.960483989341936, 'reg__subsample': 0.8982357554484275, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7305809087206884, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9983.193176782122}

Rank 7: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.0162599671556961, 'reg__l2_leaf_reg': 17.527743963435633, 'reg__bagging_temperature': 0.31516450210312463, 'reg__random_strength': 2.7157691346097135, 'reg__subsample': 0.5245326448640045,

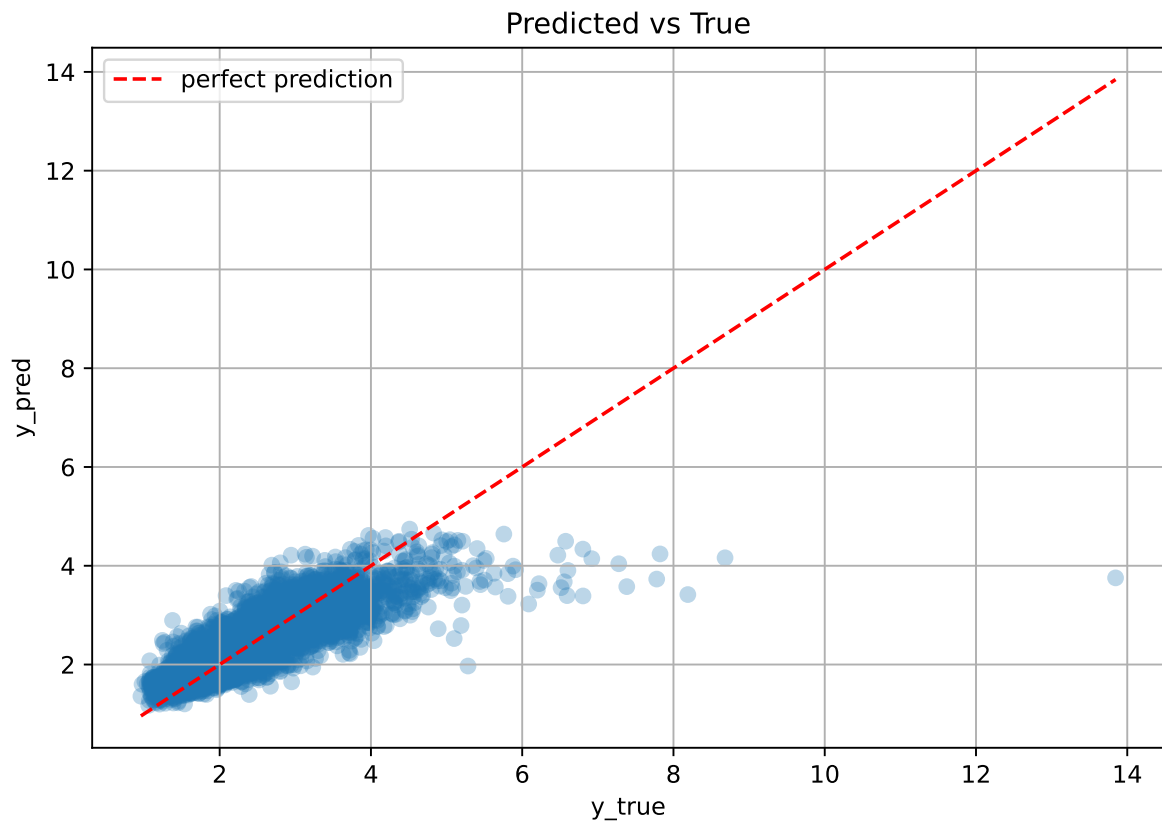
'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8452257037636954, 'reg__border_count': 77, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1530.3853854754434}

Rank 8: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.02468907807552143, 'reg__l2_leaf_reg': 32.58267837558499, 'reg__bagging_temperature': 4.0106697023036455, 'reg__random_strength': 3.421693529948888, 'reg__subsample': 0.6913070018246654, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7199376713623458, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1130.4323697414466}

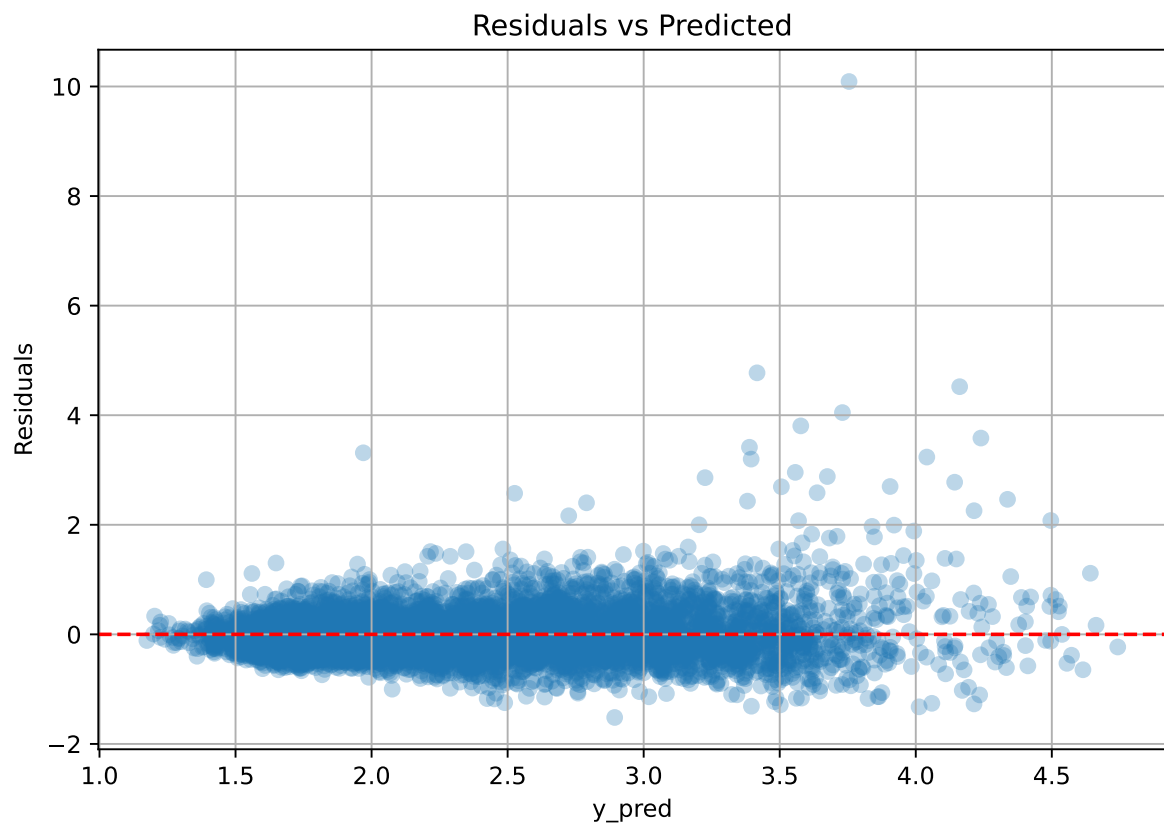
Rank 9: {'reg__iterations': 1000, 'reg__depth': 4, 'reg__learning_rate': 0.012195450656013525, 'reg__l2_leaf_reg': 37.256653341846075, 'reg__bagging_temperature': 0.640098260279488, 'reg__random_strength': 3.23272778711388, 'reg__subsample': 0.7666589766190806, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5185105073077442, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5401.812926171988}

Rank 10: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.01388210341168158, 'reg__l2_leaf_reg': 43.56012535876318, 'reg__bagging_temperature': 0.3340931270010342, 'reg__random_strength': 2.7165213778289656, 'reg__subsample': 0.7543799587132468, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.5668606954268132, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 6714.186721344695}

3.6.2 Scatter



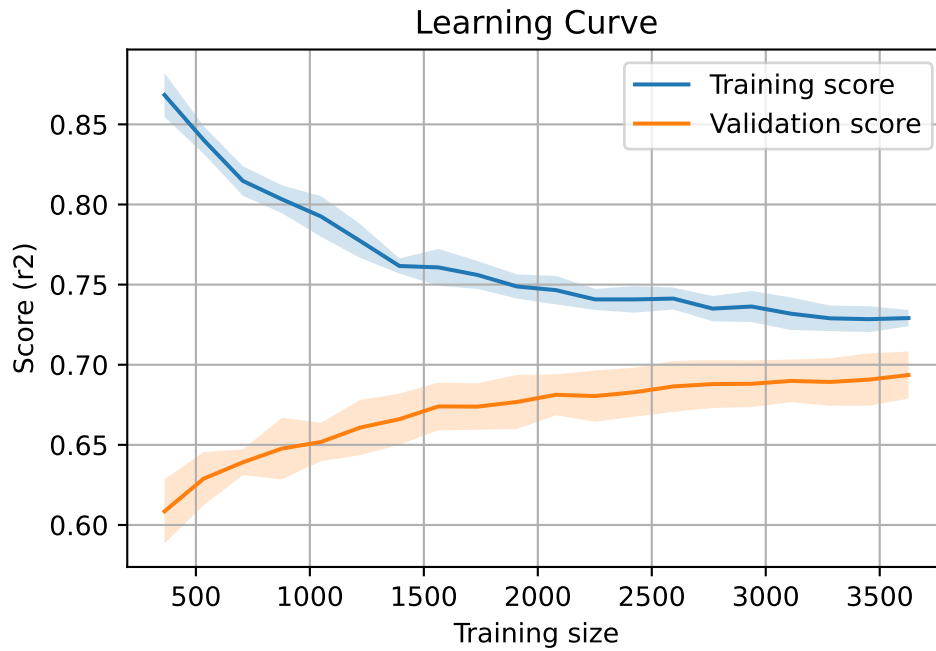
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7402, `Len(training)` : 5182,

`Len(training_real)` : 3628, `Len(test_of_training)` : 1554, `Len(test)` : 2220



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.533	0.529	0.014	0.509	0.509	-0.02	0.963	1.008	nan
LGBM	0.686	0.656	0.015	0.612	0.635	-0.021	0.968	1.045	0.6558
CatBoost	0.728	0.693	0.016	0.644	0.651	-0.042	0.94	1.05	0.6933

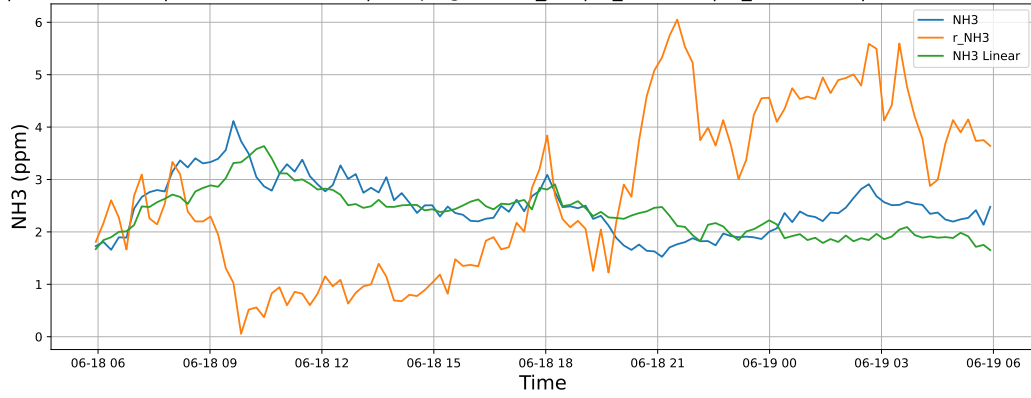
4 Plot

4.1 Standalone plots

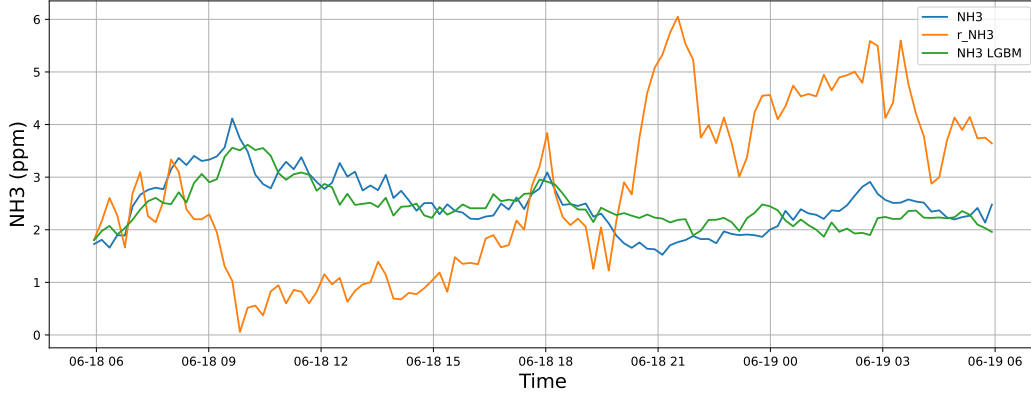
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

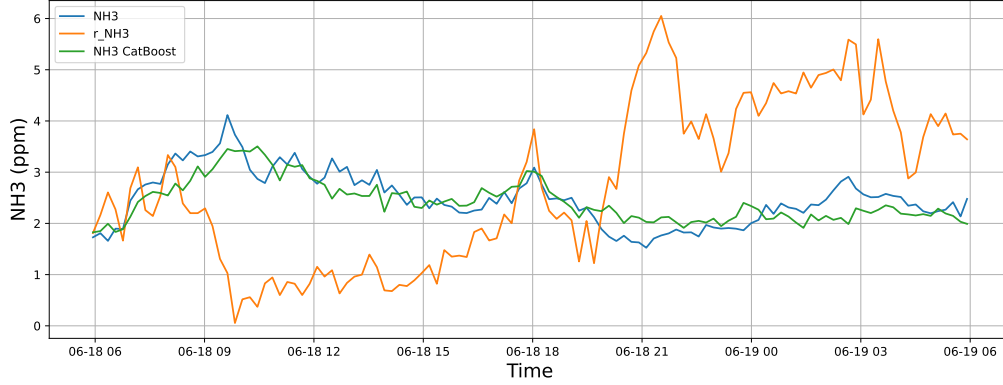
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

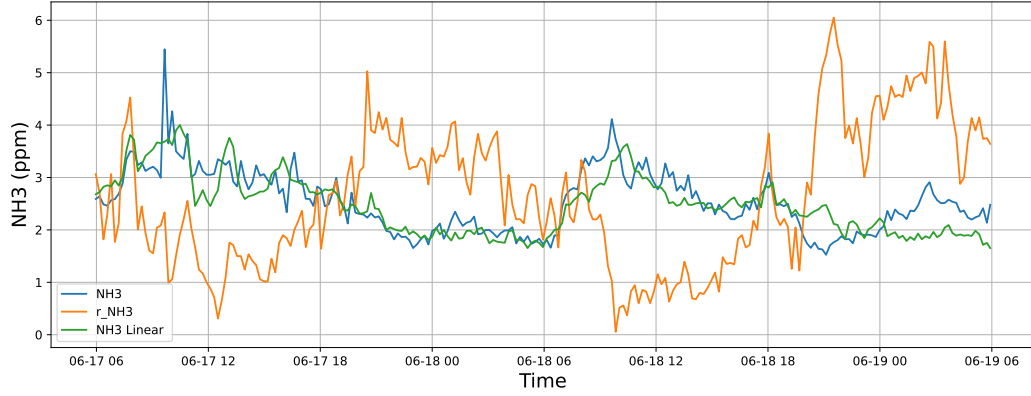


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

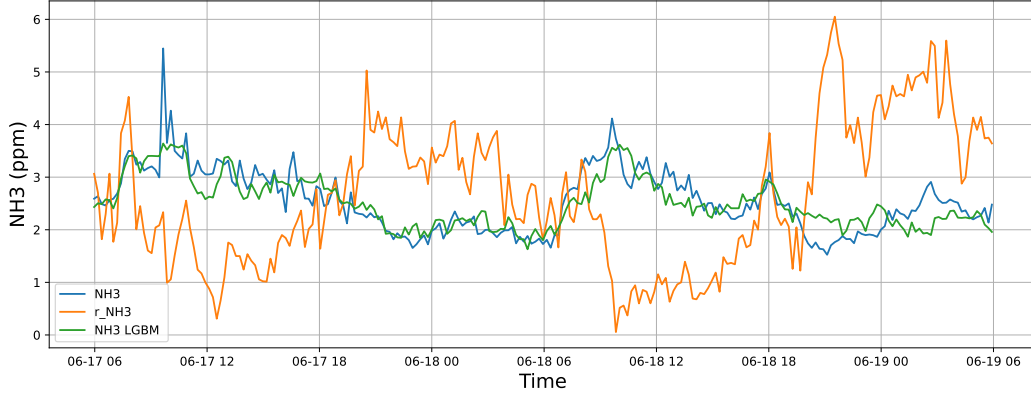


4.1.1.2 Time window : 48

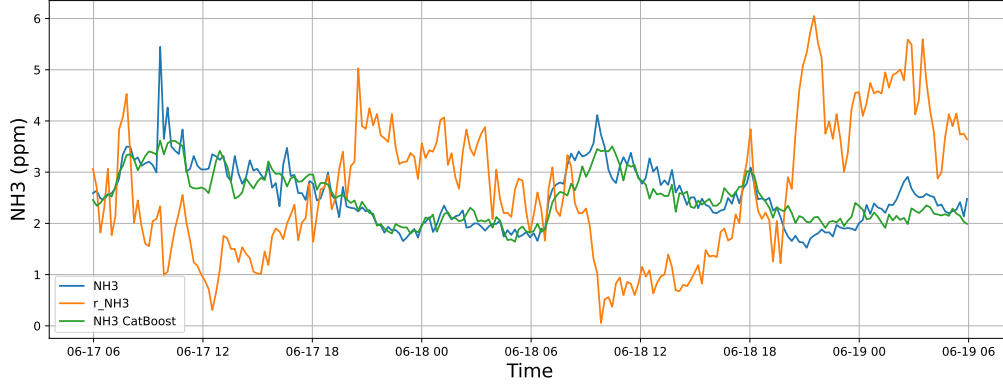
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

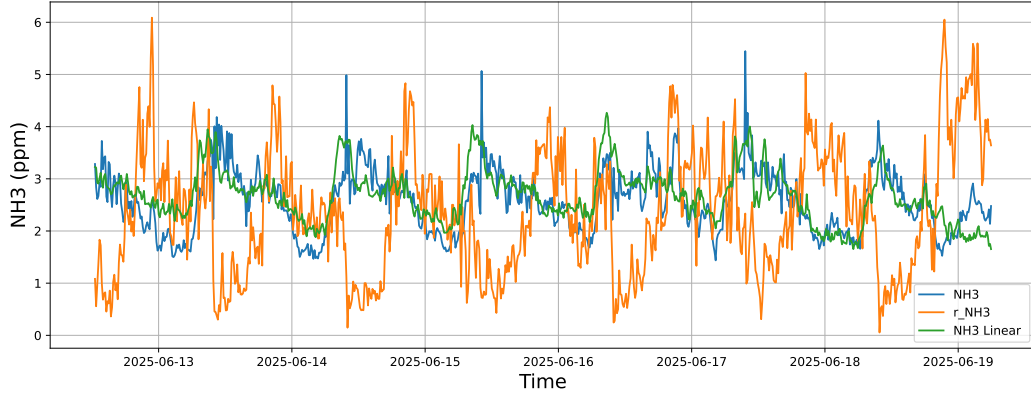


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

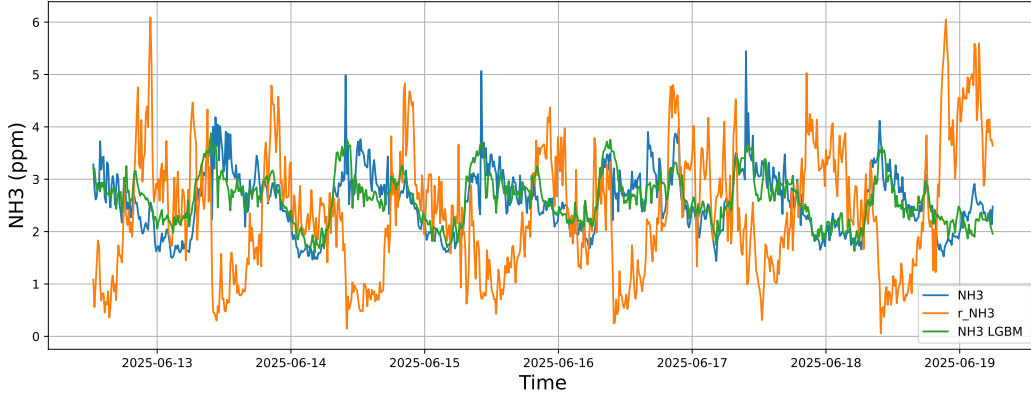


4.1.1.3 Time window : ALL

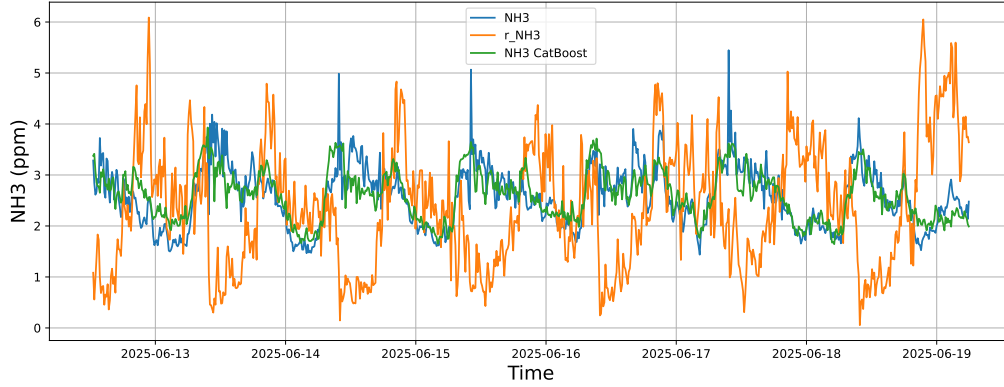
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



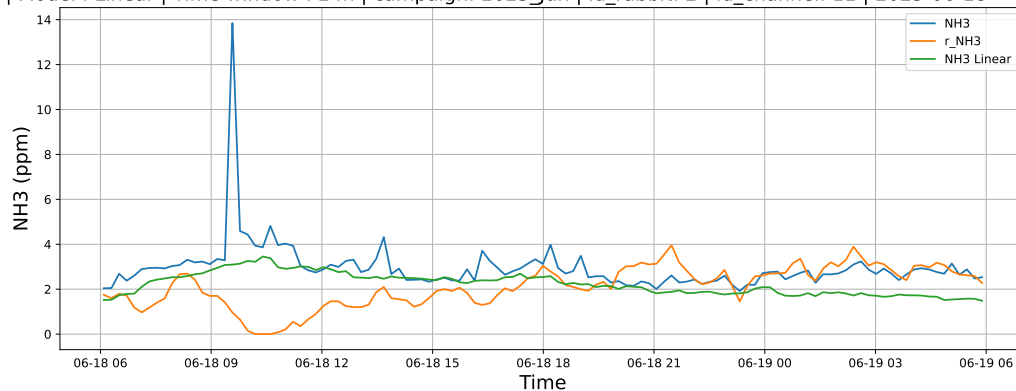
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



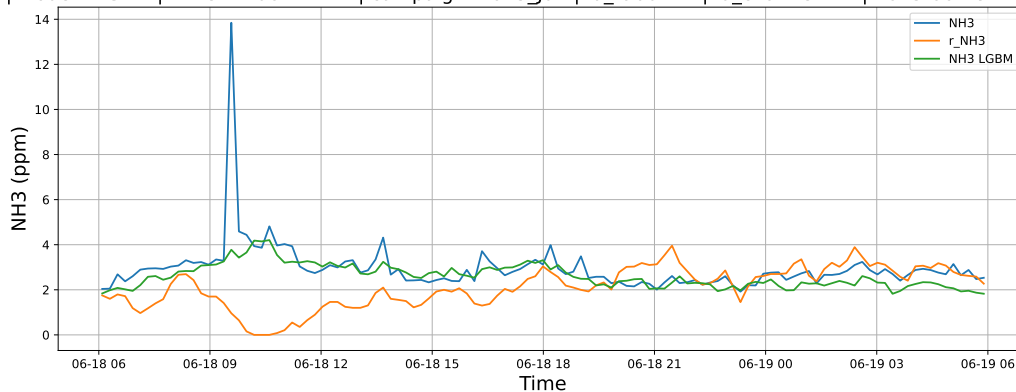
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

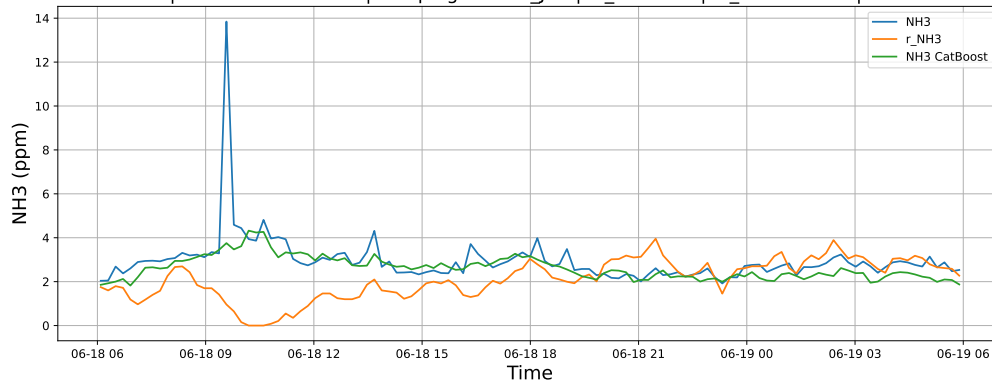
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

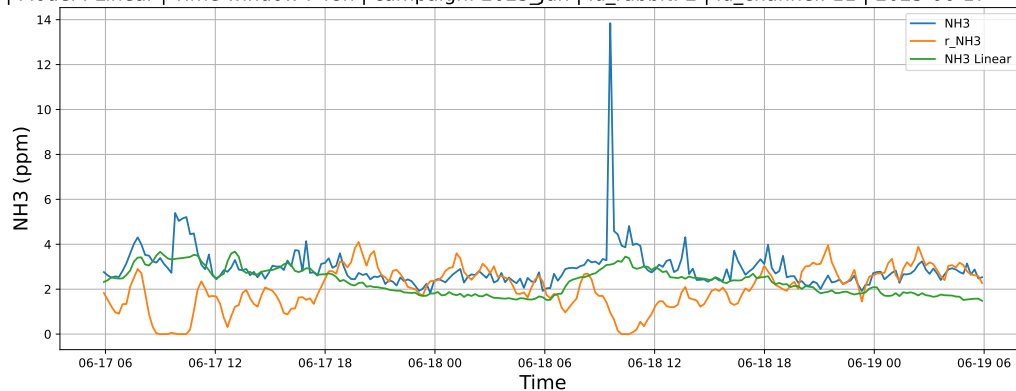


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

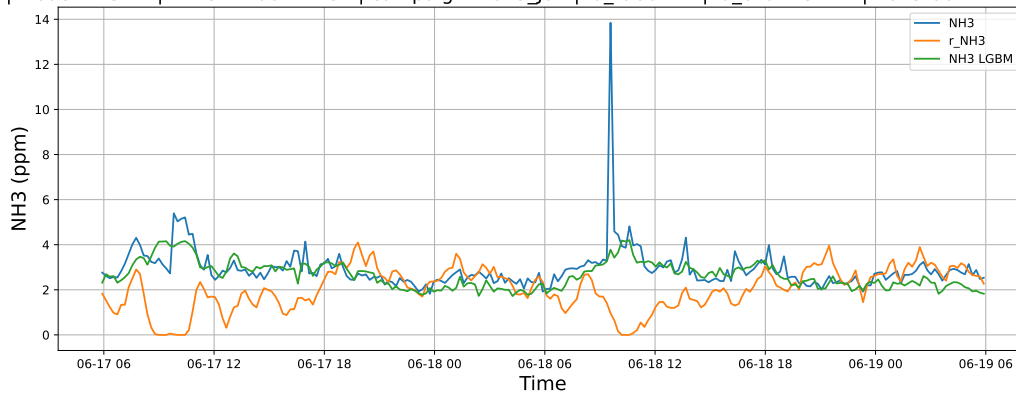


4.1.2.2 Time window : 48

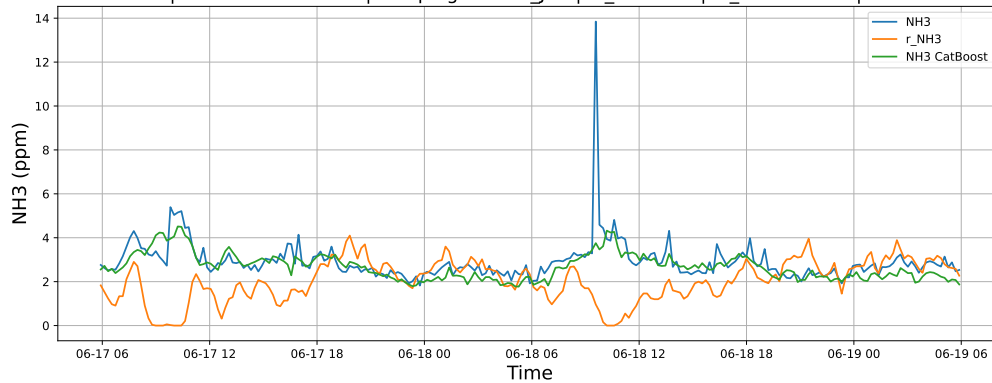
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

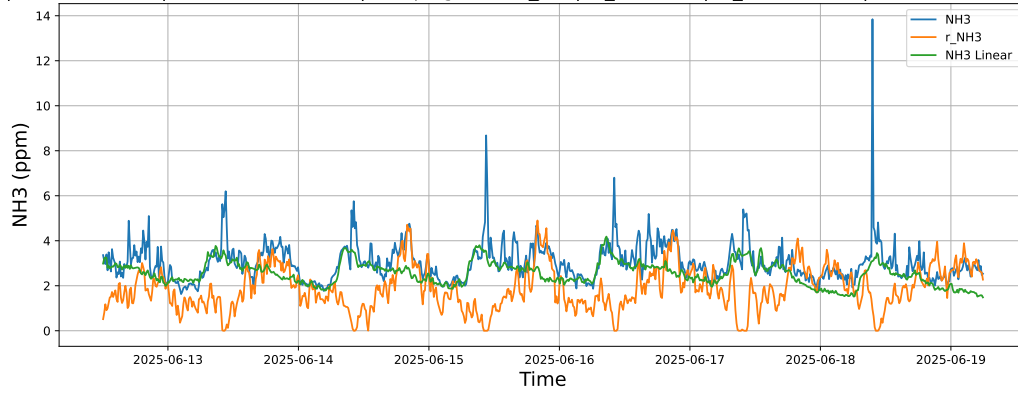


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

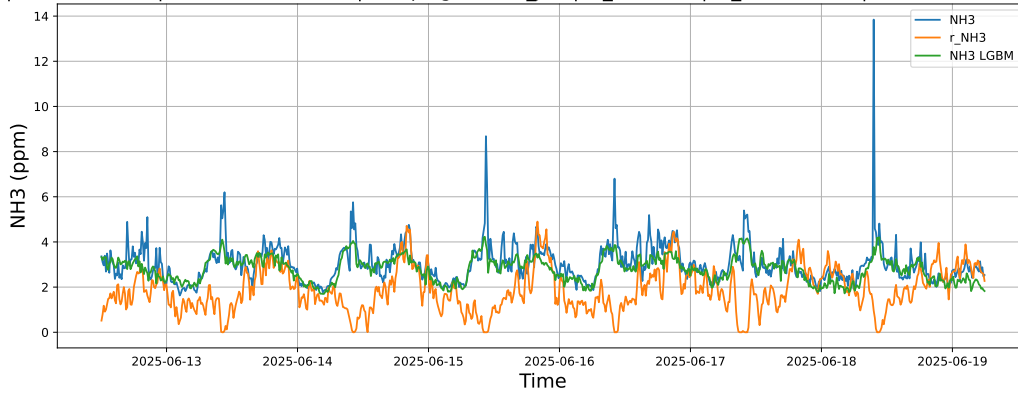


4.1.2.3 Time window : ALL

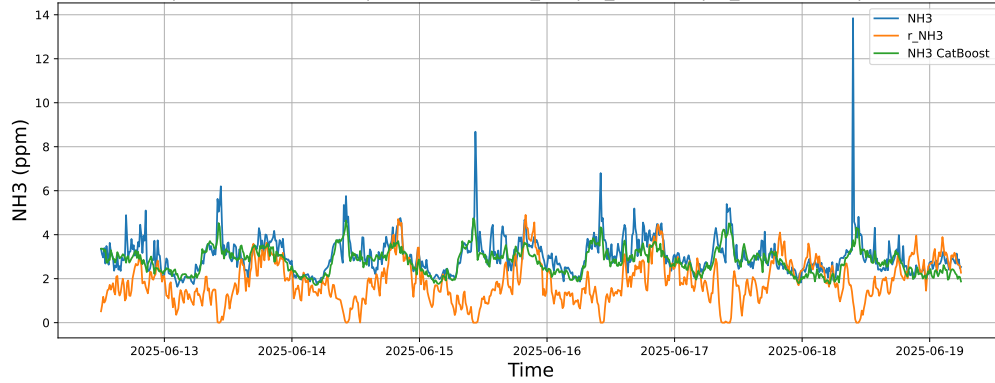
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



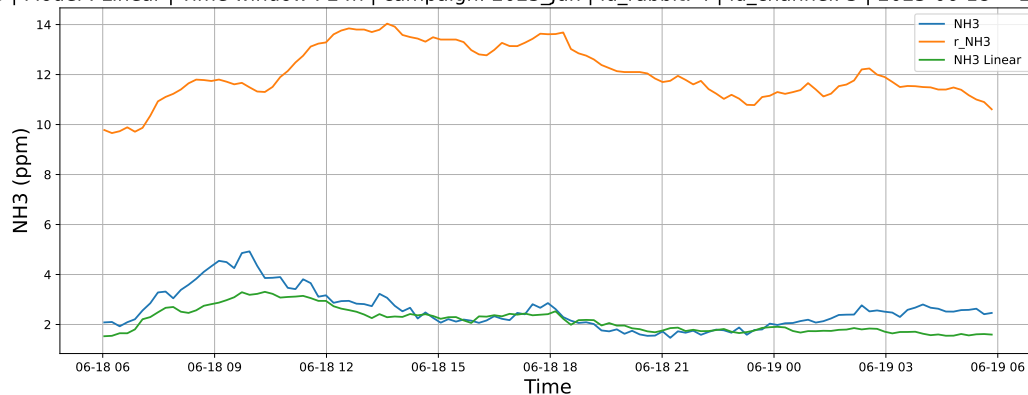
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



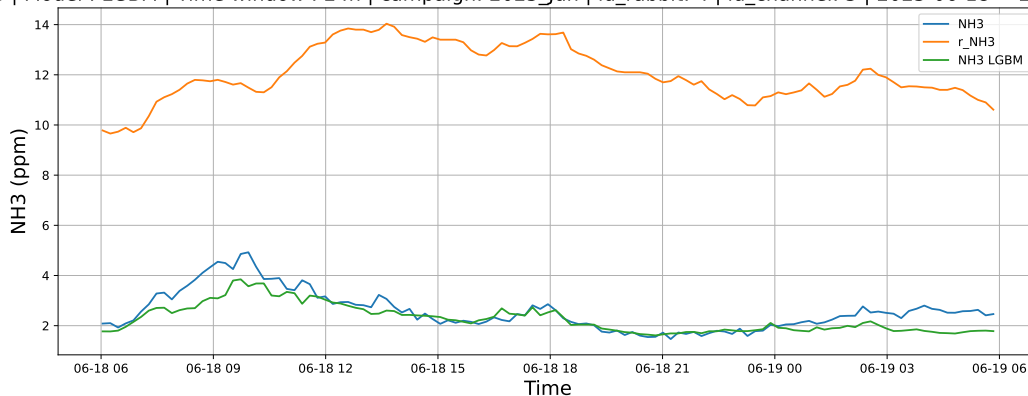
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

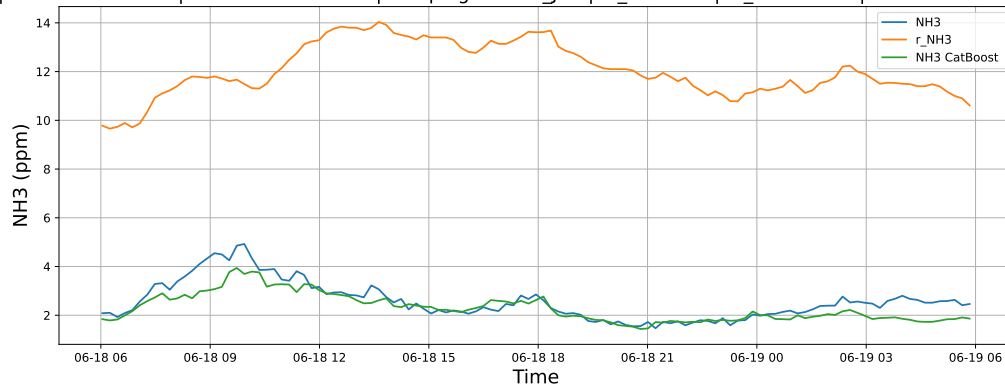
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

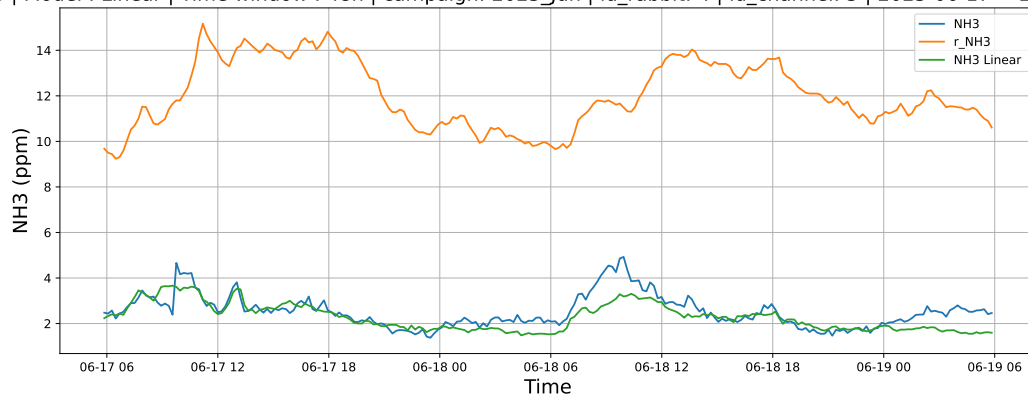


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

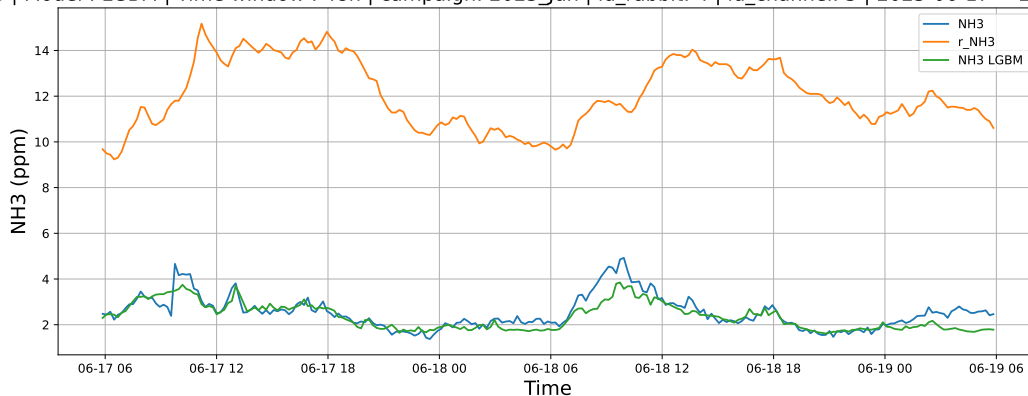


4.1.3.2 Time window : 48

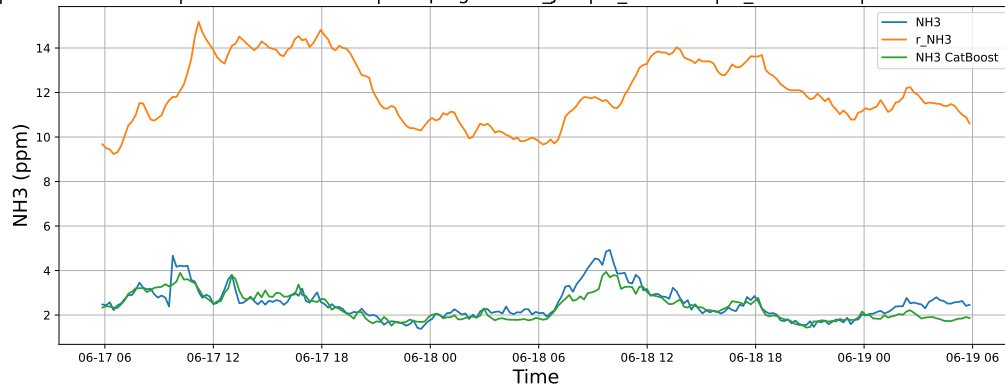
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

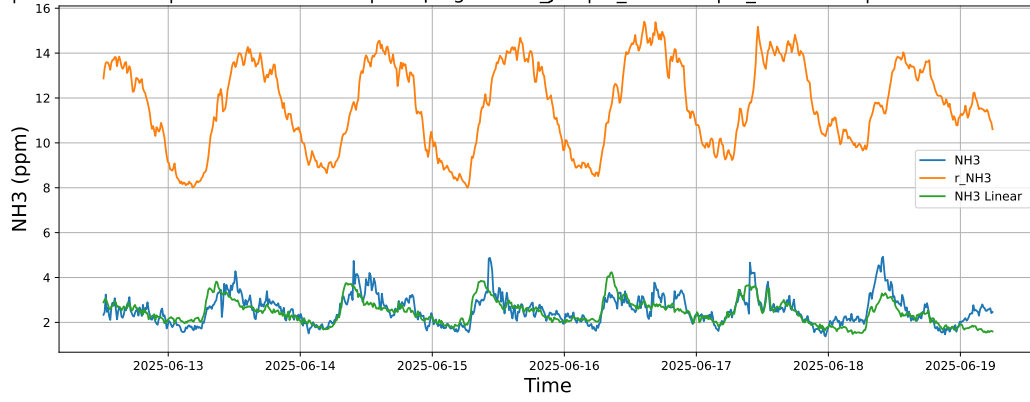


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

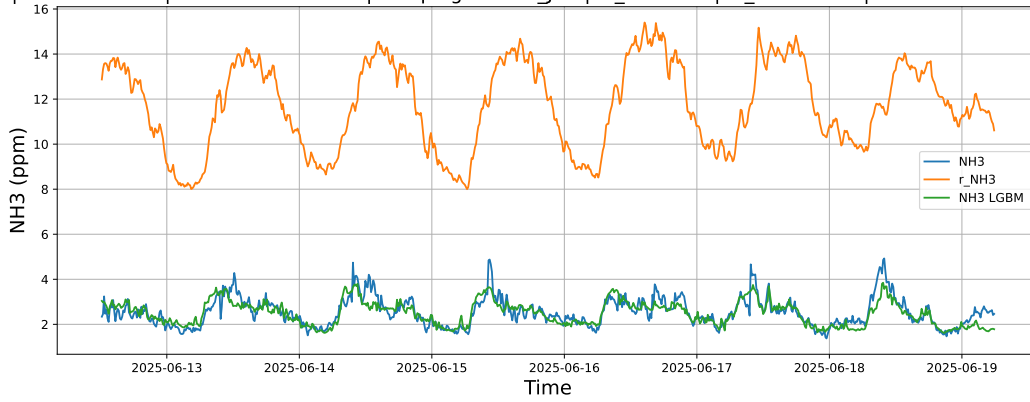


4.1.3.3 Time window : ALL

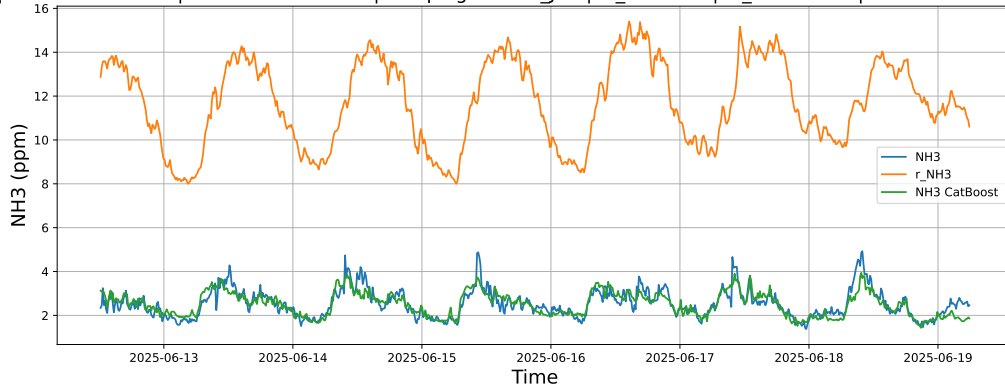
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



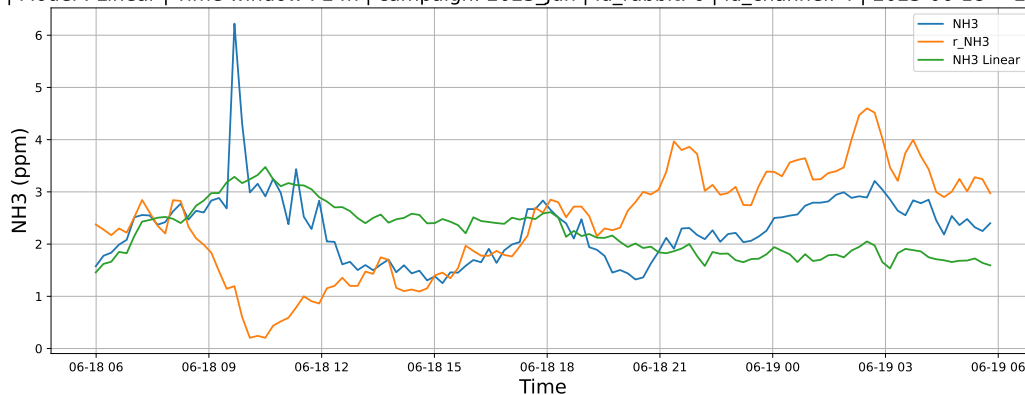
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



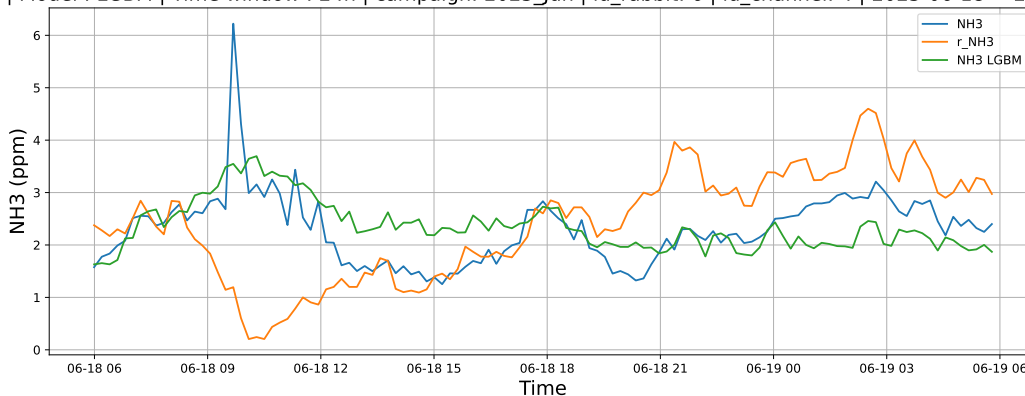
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

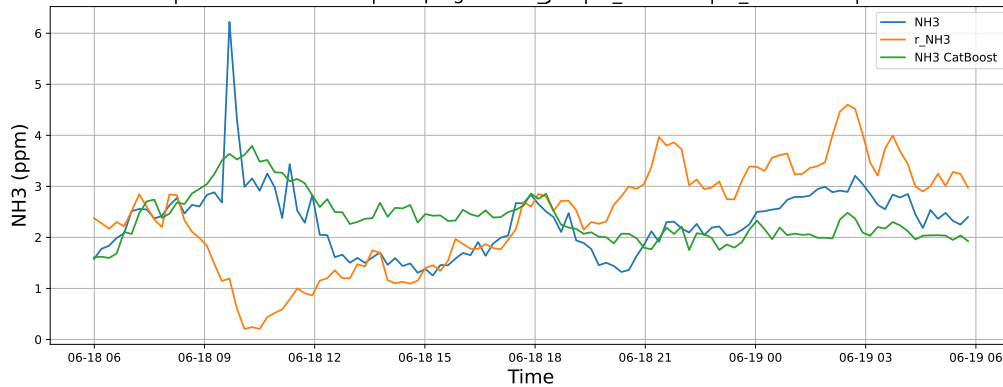
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

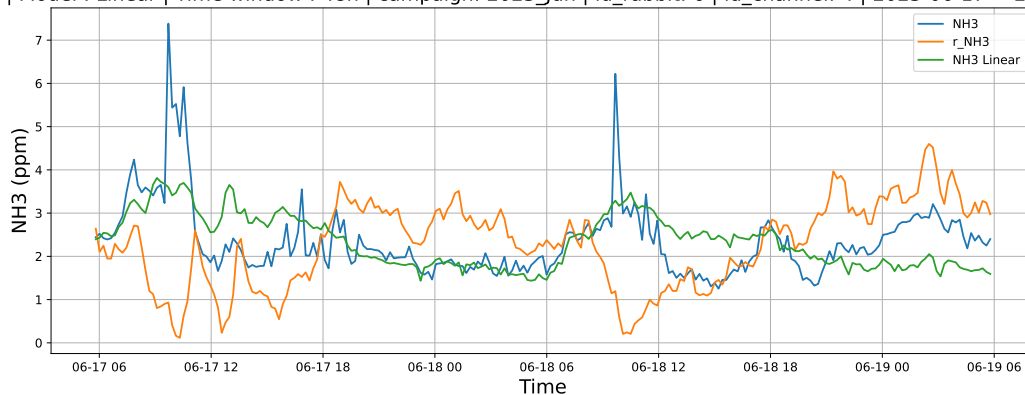


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

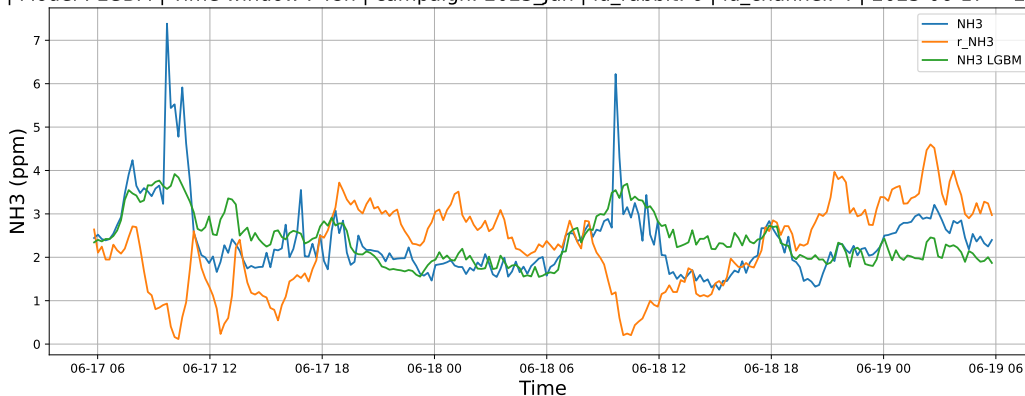


4.1.4.2 Time window : 48

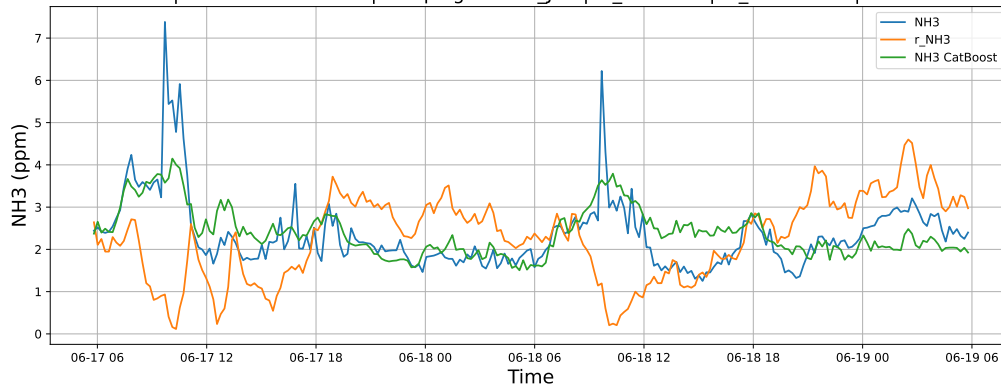
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

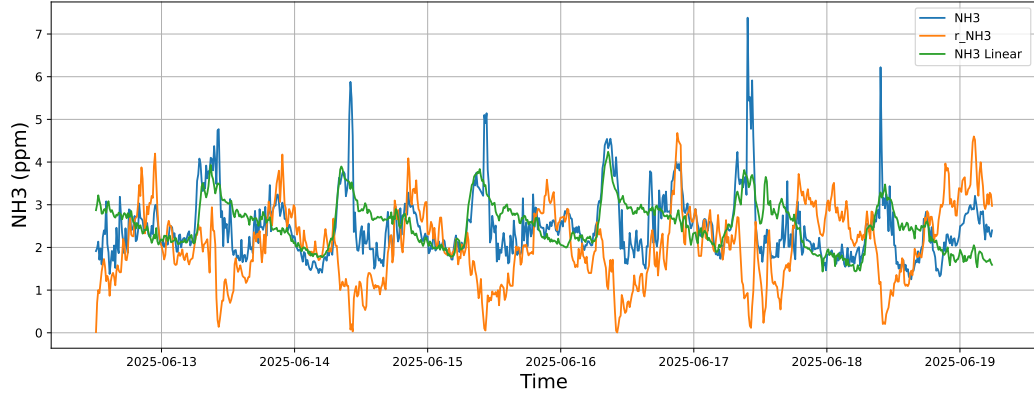


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

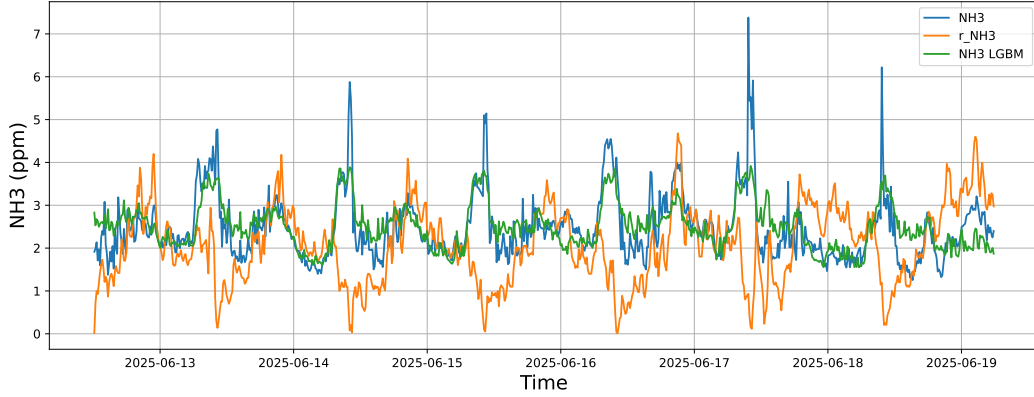


4.1.4.3 Time window : ALL

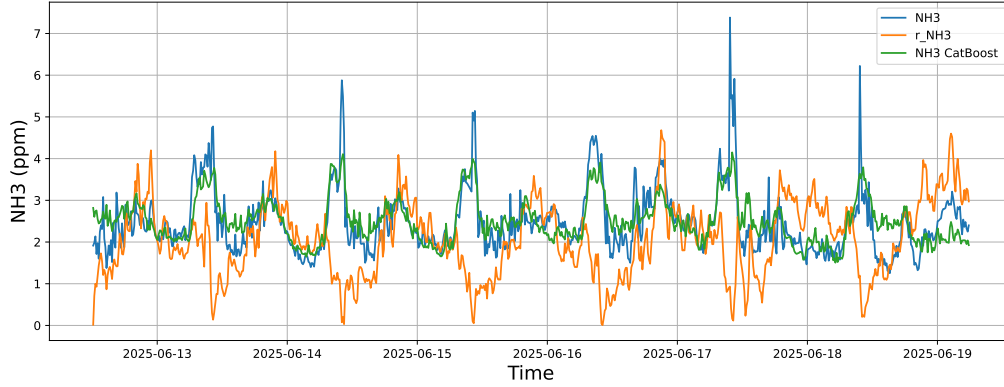
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



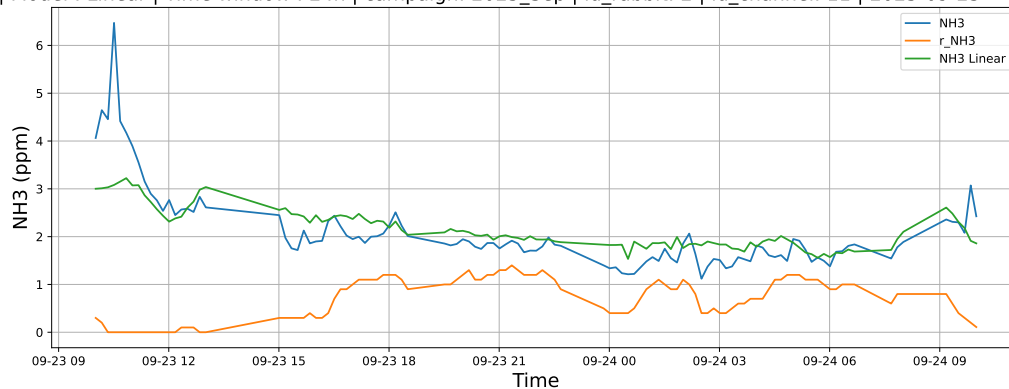
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



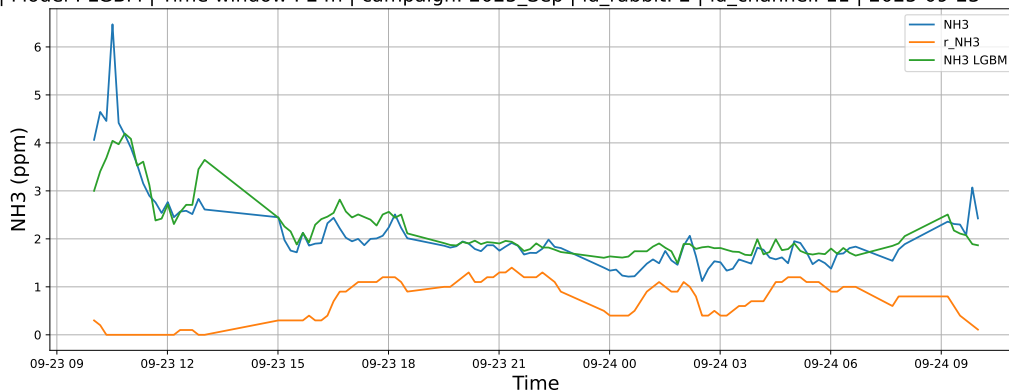
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

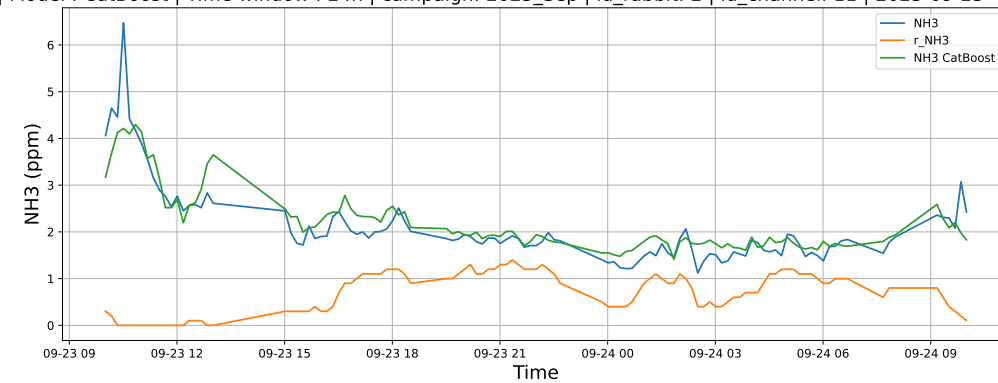
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

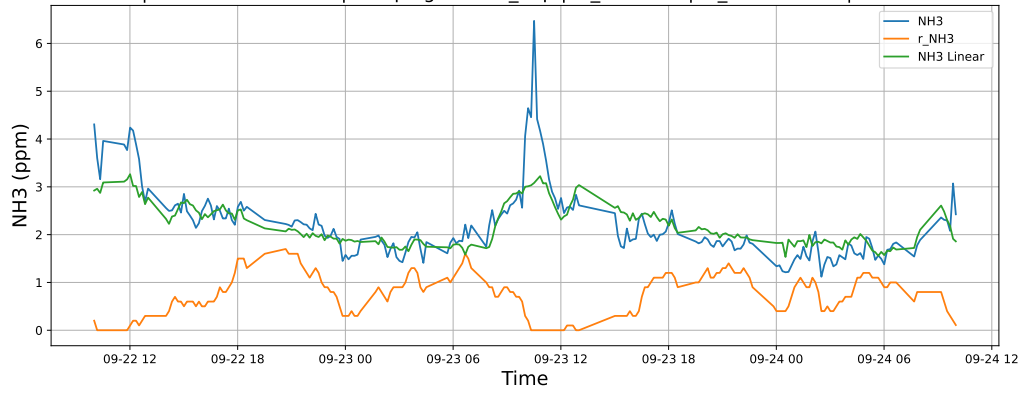


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

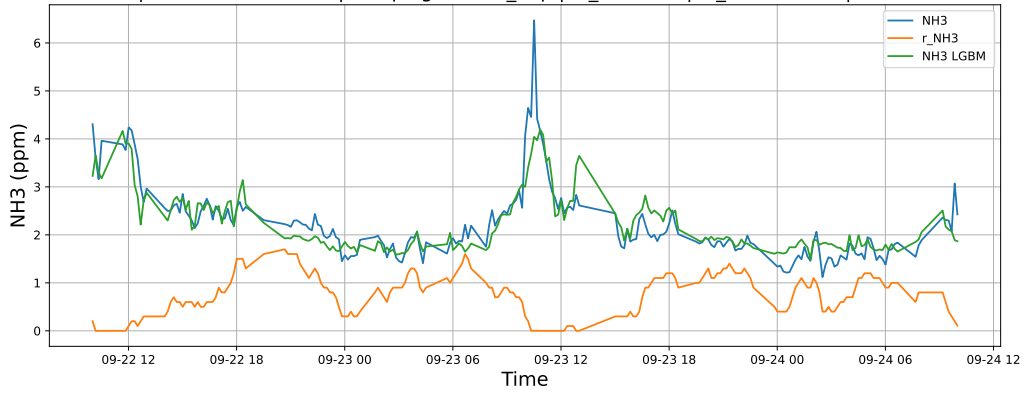


4.1.5.2 Time window : 48

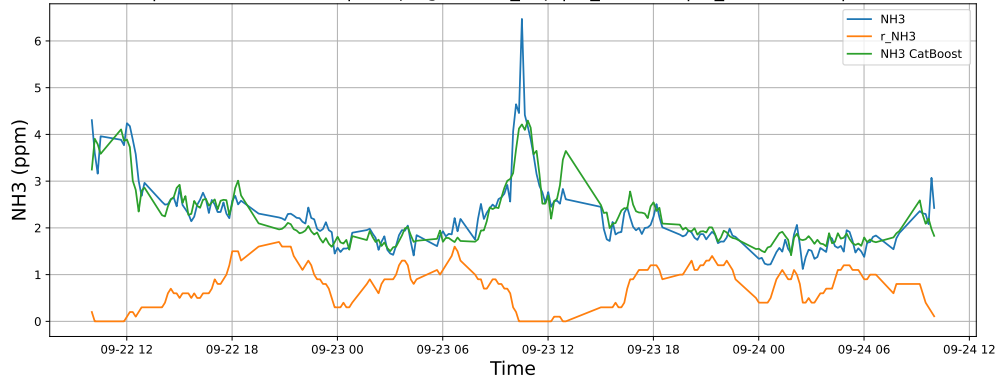
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

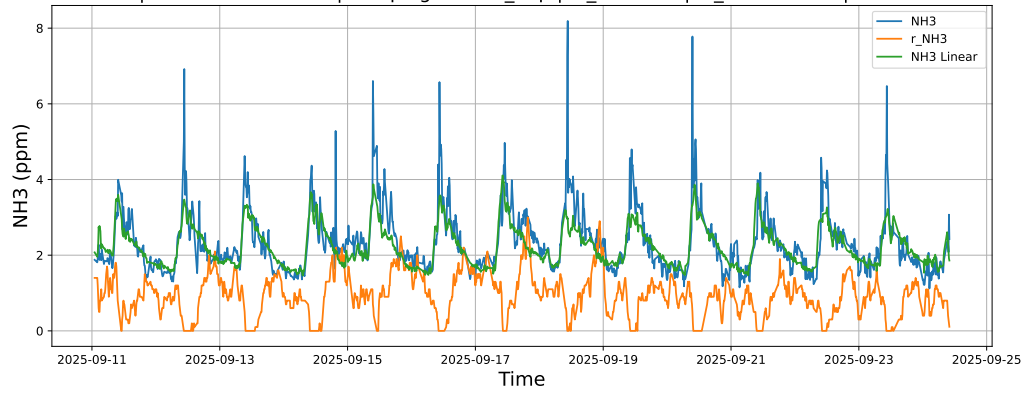


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

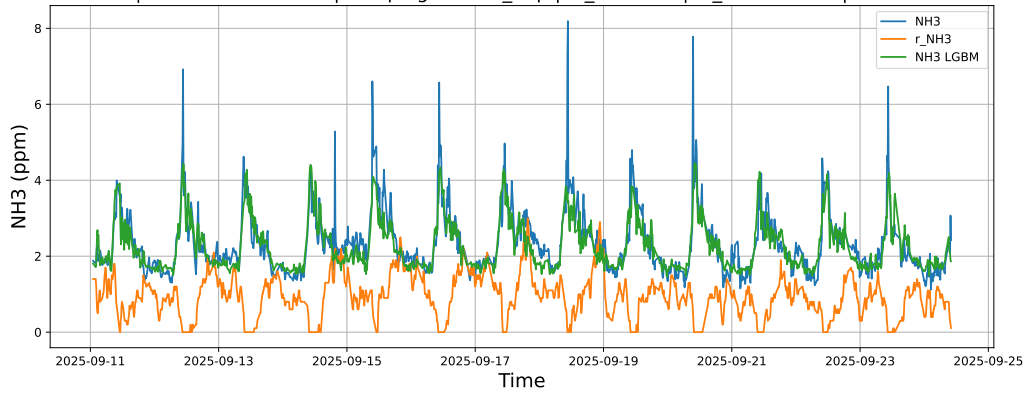


4.1.5.3 Time window : ALL

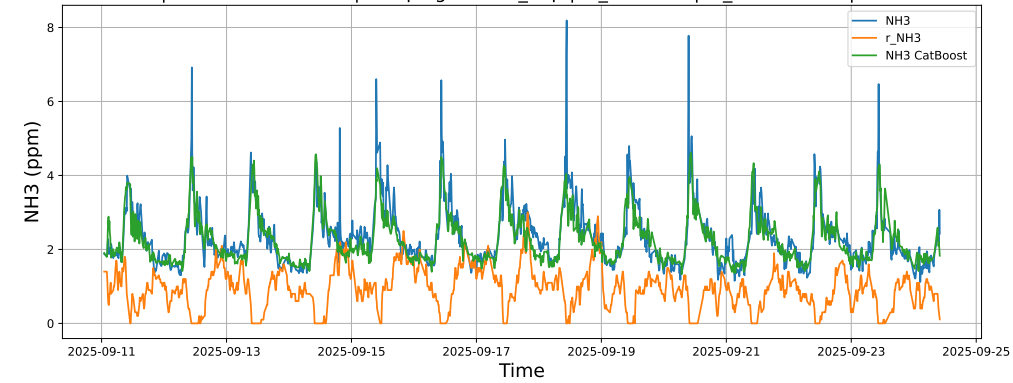
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



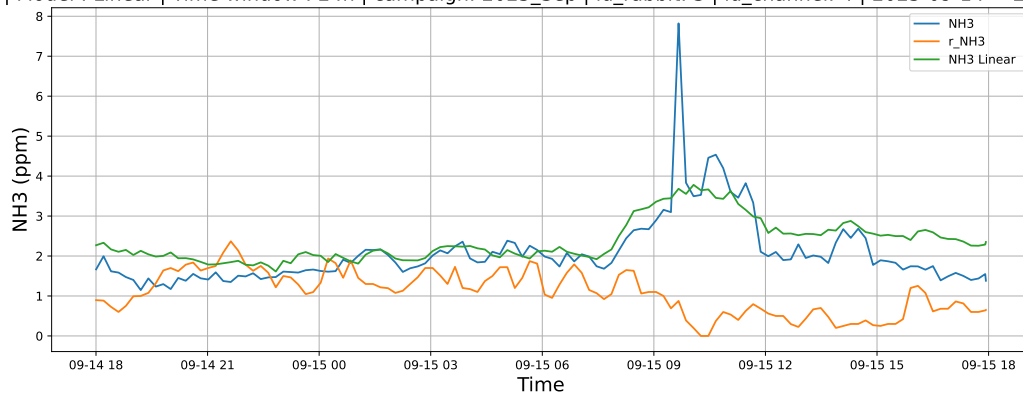
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



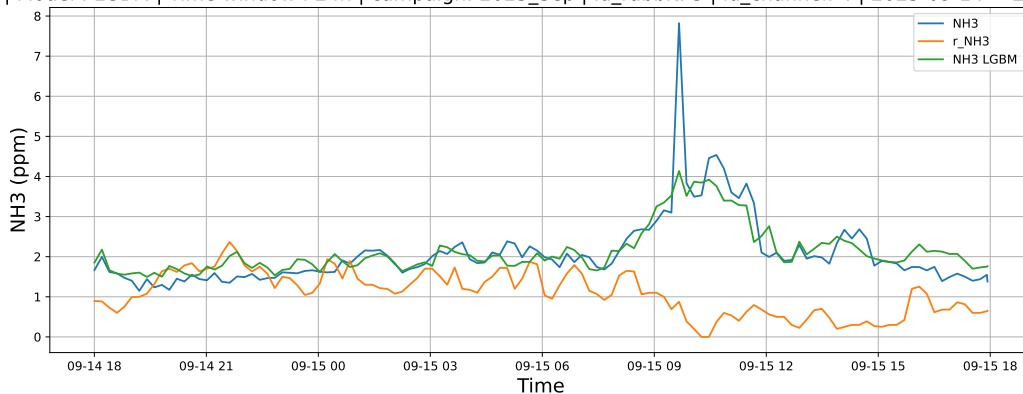
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

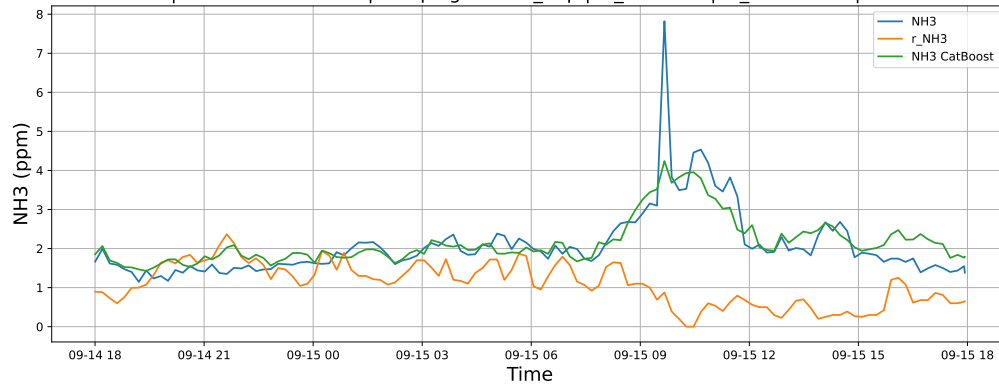
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

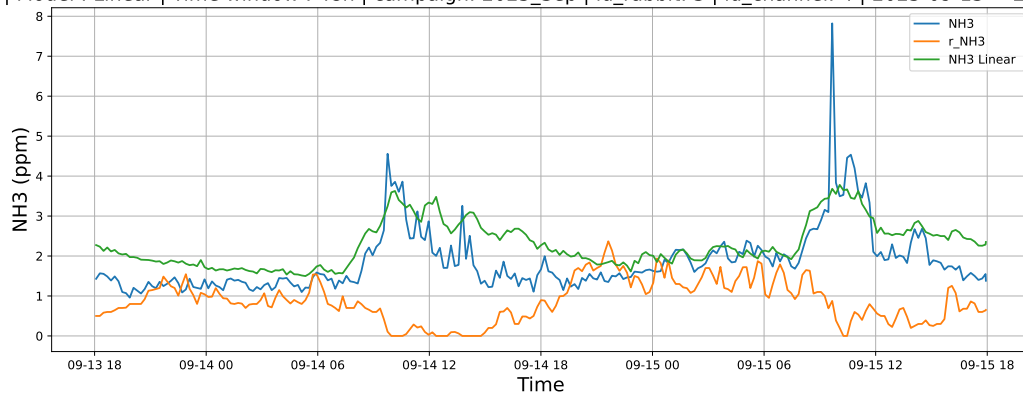


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

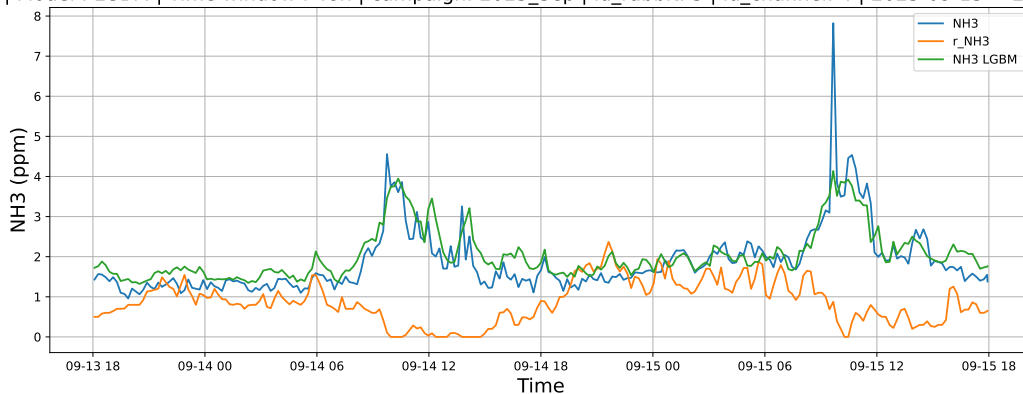


4.1.6.2 Time window : 48

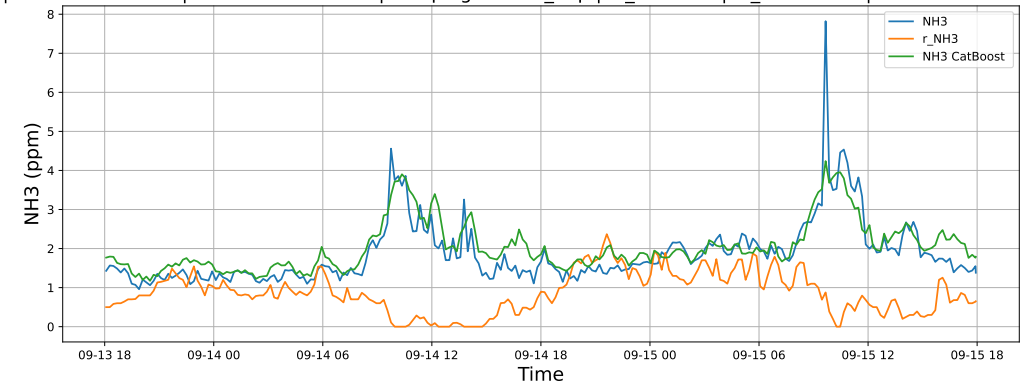
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

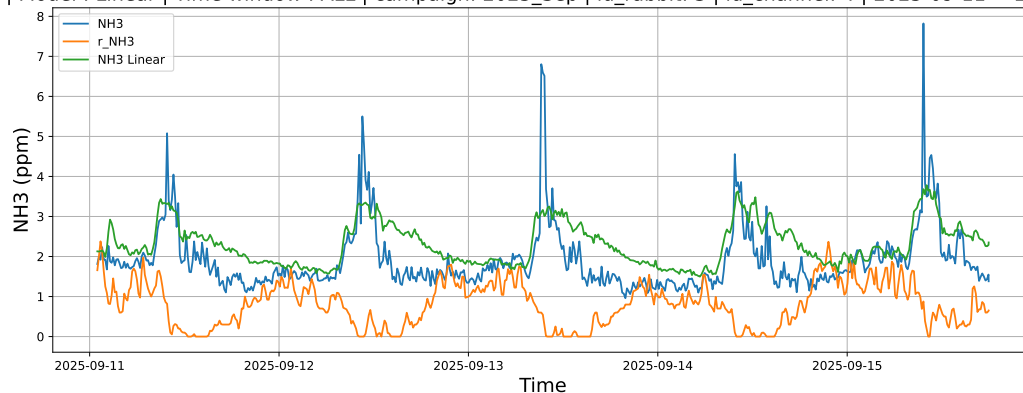


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

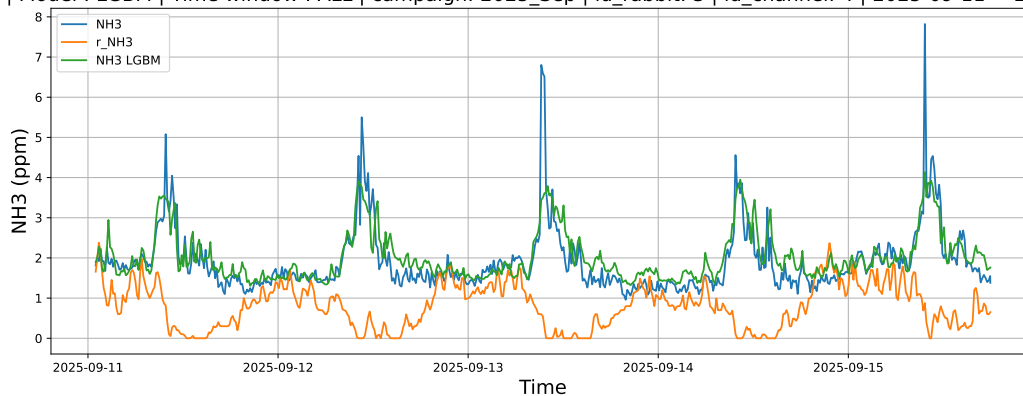


4.1.6.3 Time window : ALL

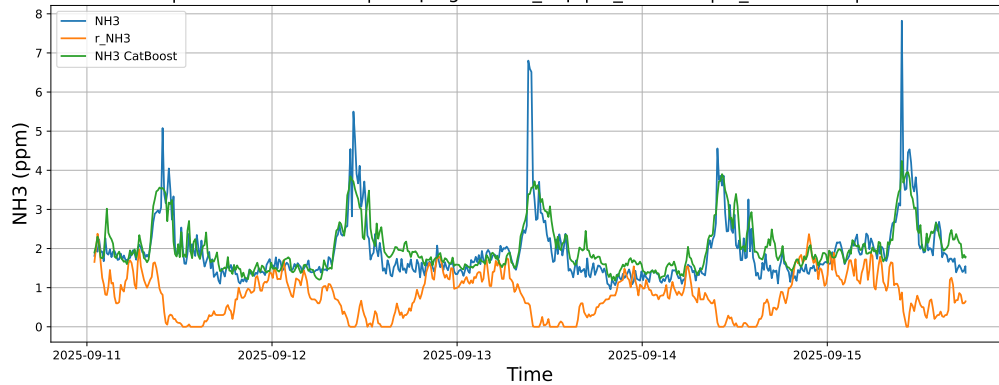
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



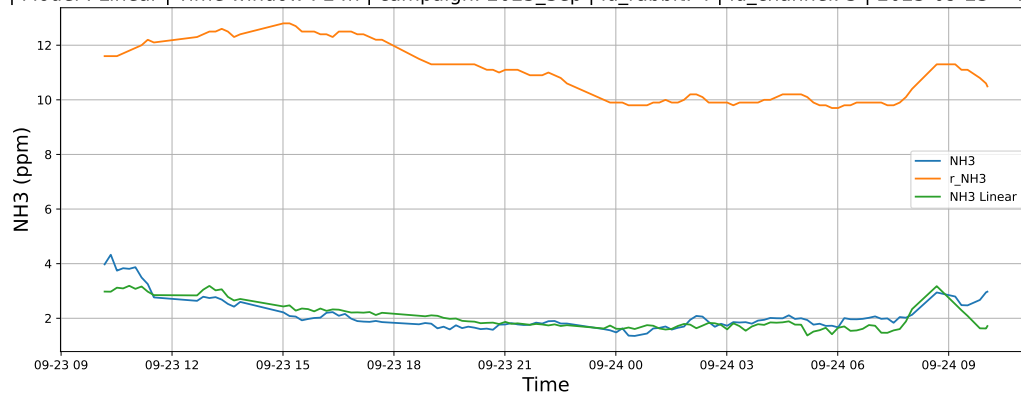
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



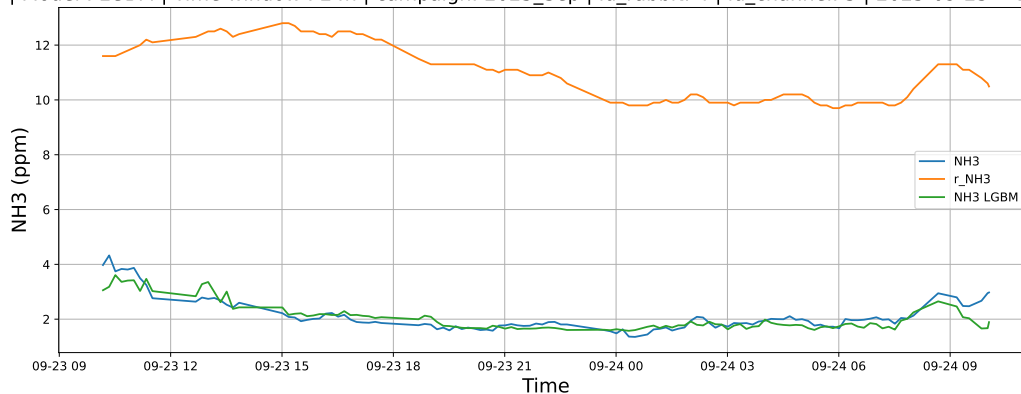
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

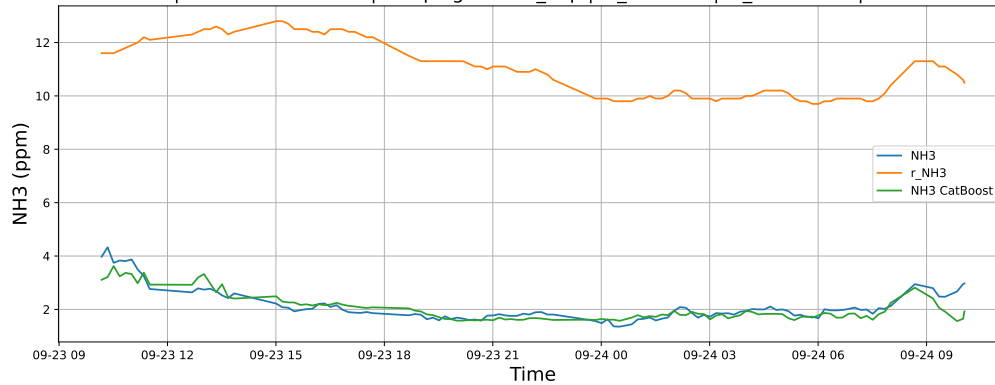
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

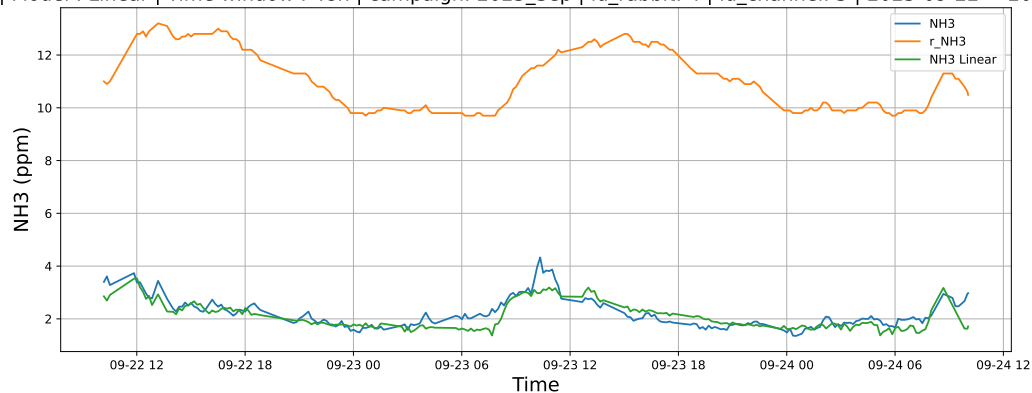


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

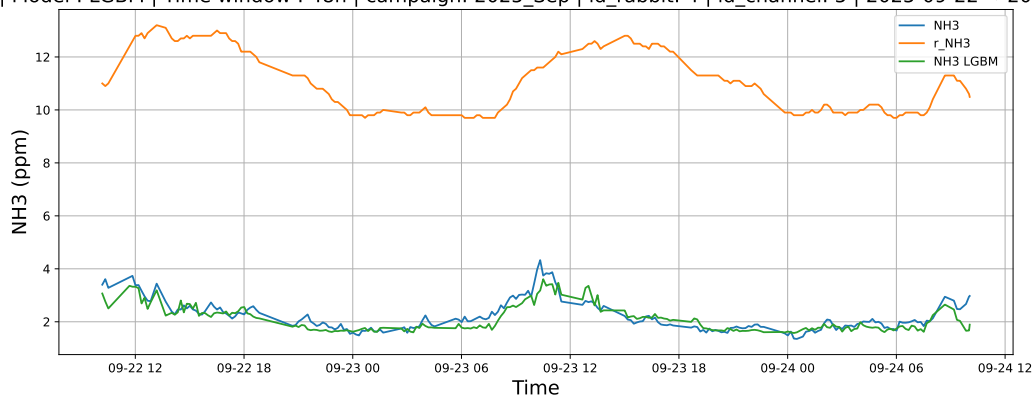


4.1.7.2 Time window : 48

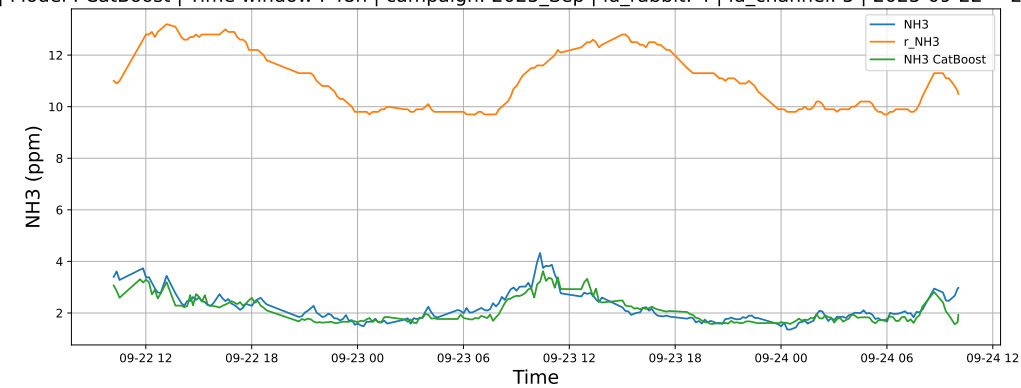
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

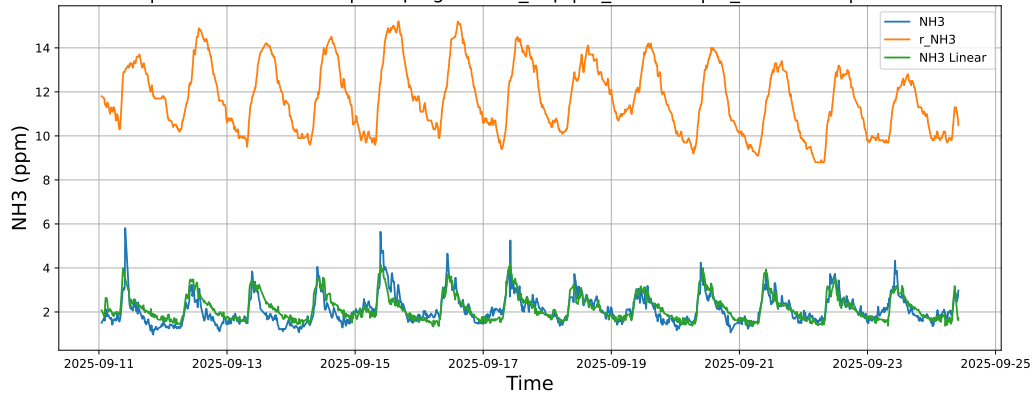


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

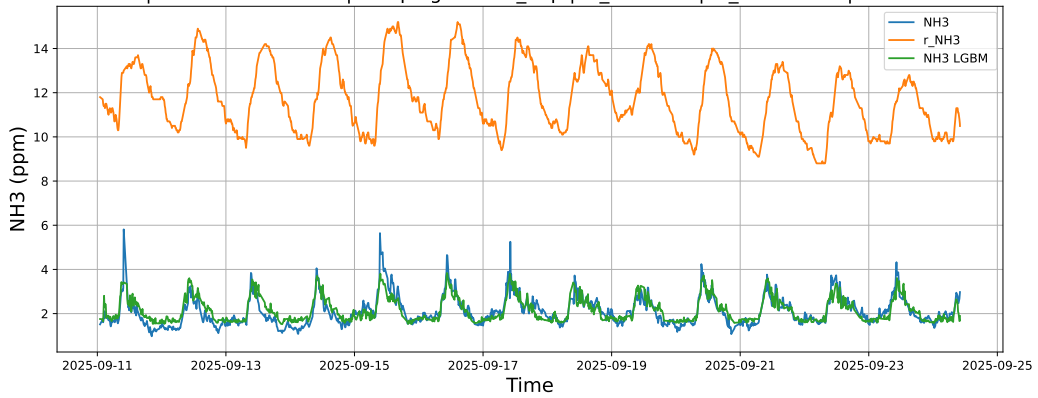


4.1.7.3 Time window : ALL

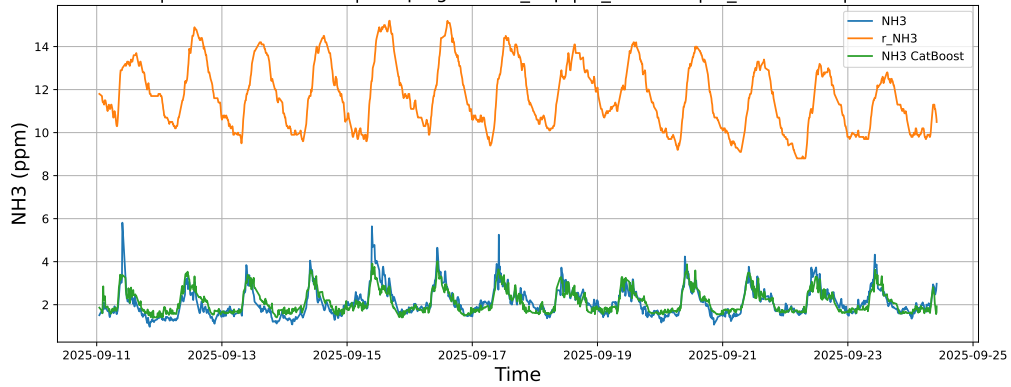
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



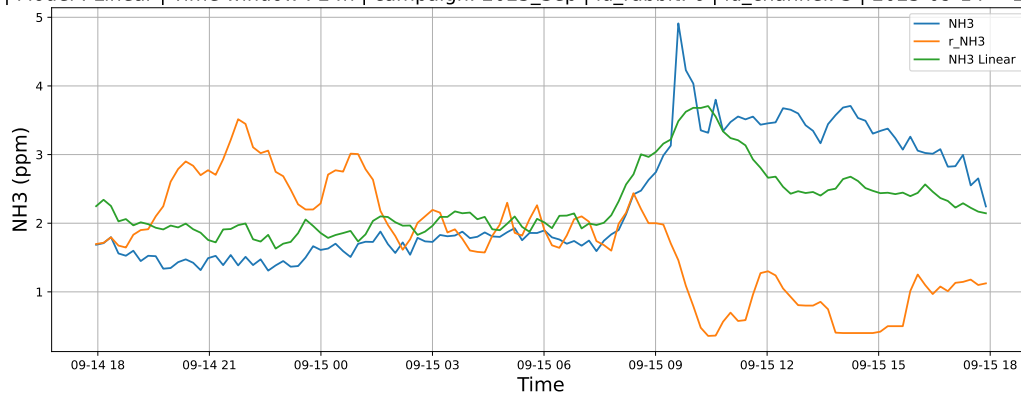
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



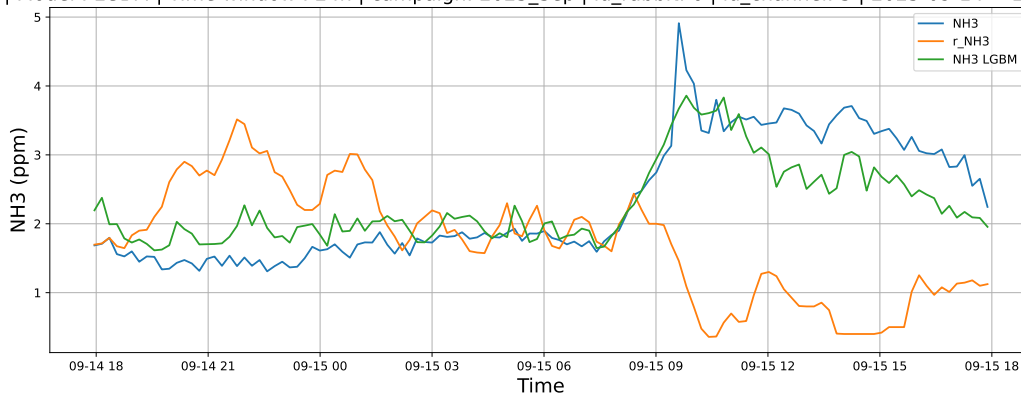
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

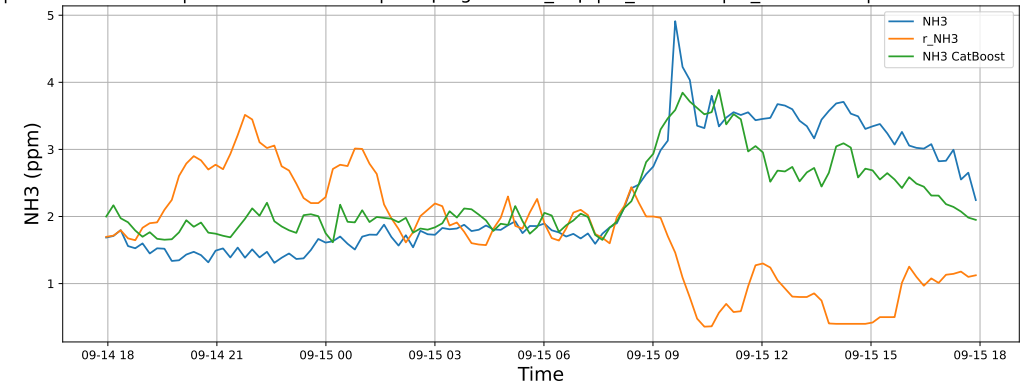
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

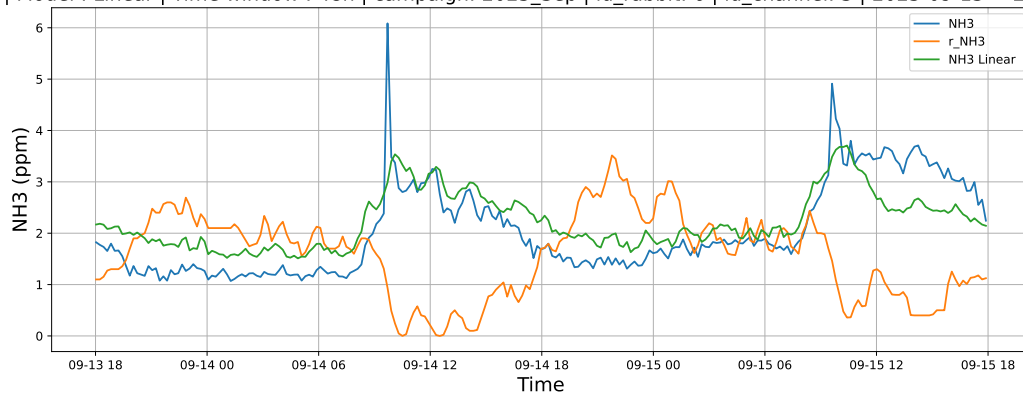


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

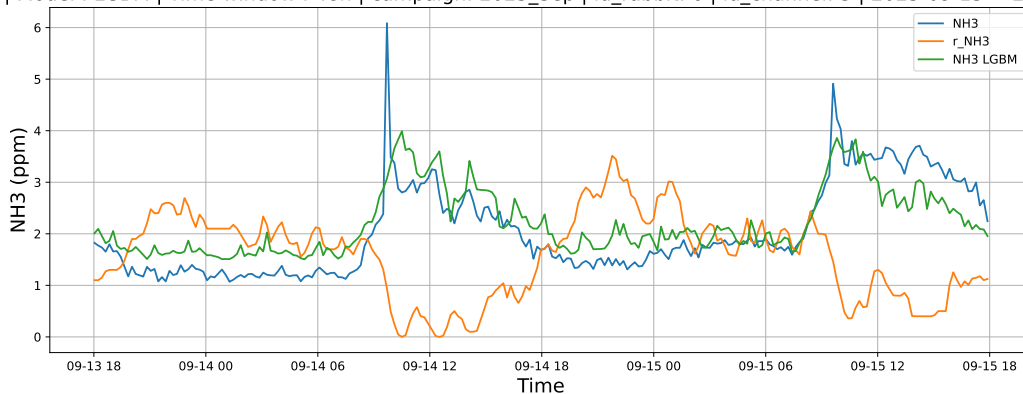


4.1.8.2 Time window : 48

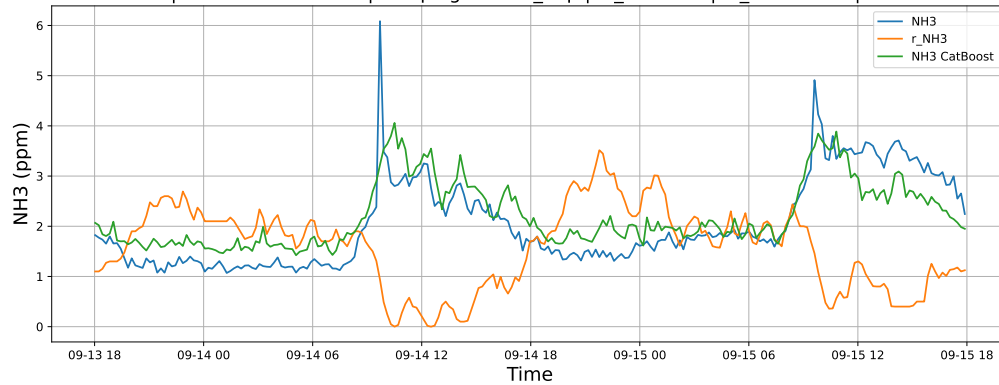
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

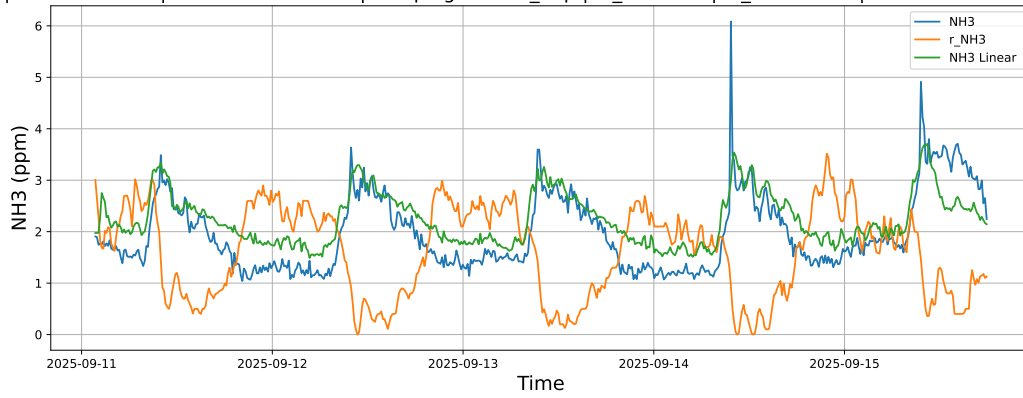


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

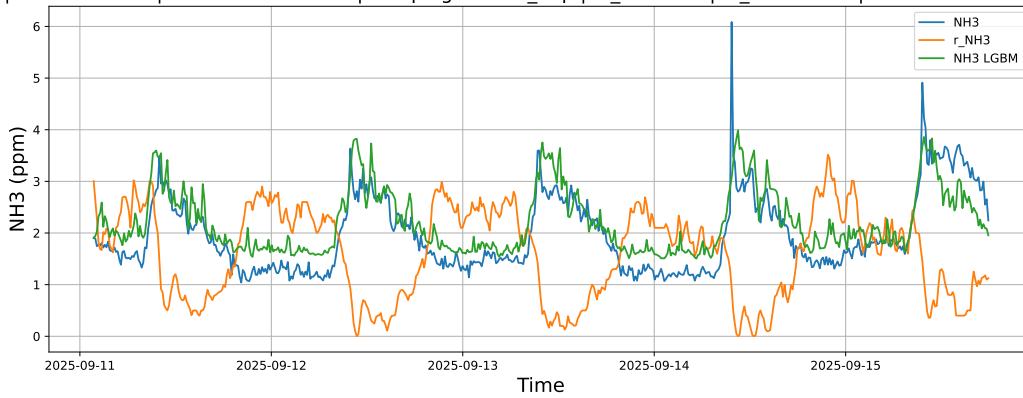


4.1.8.3 Time window : ALL

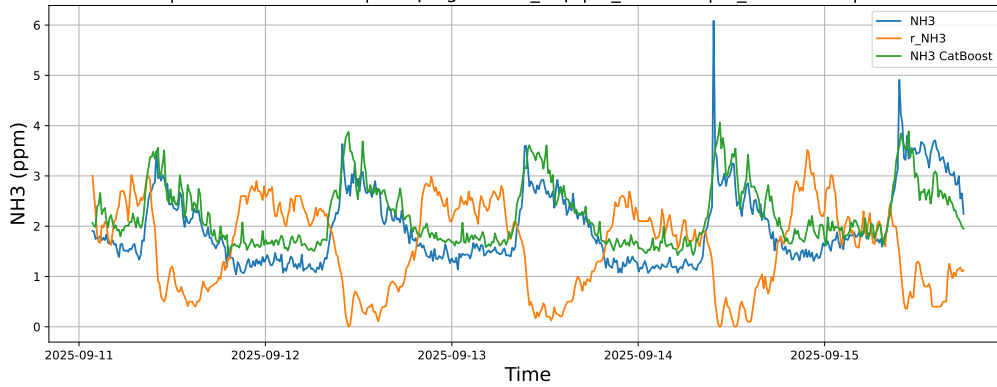
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

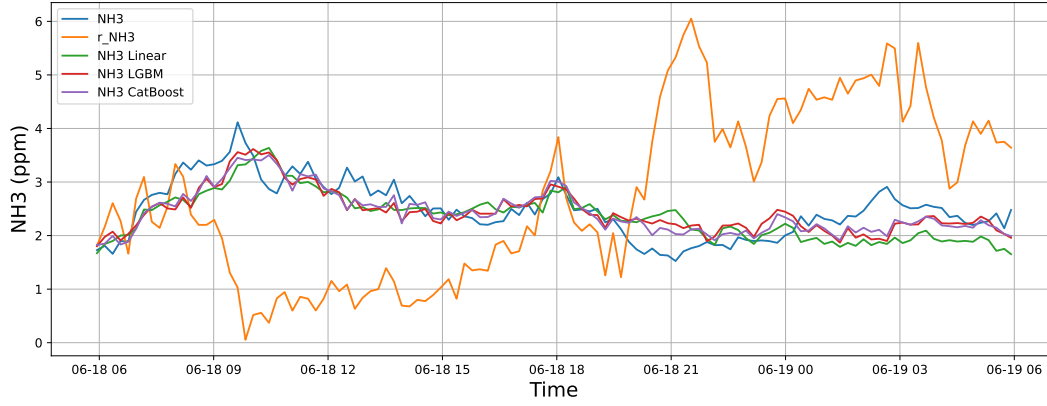


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

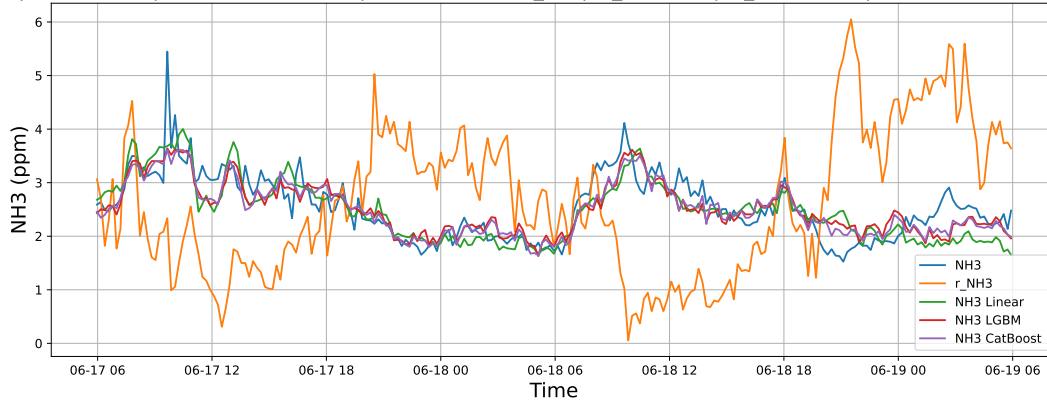
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



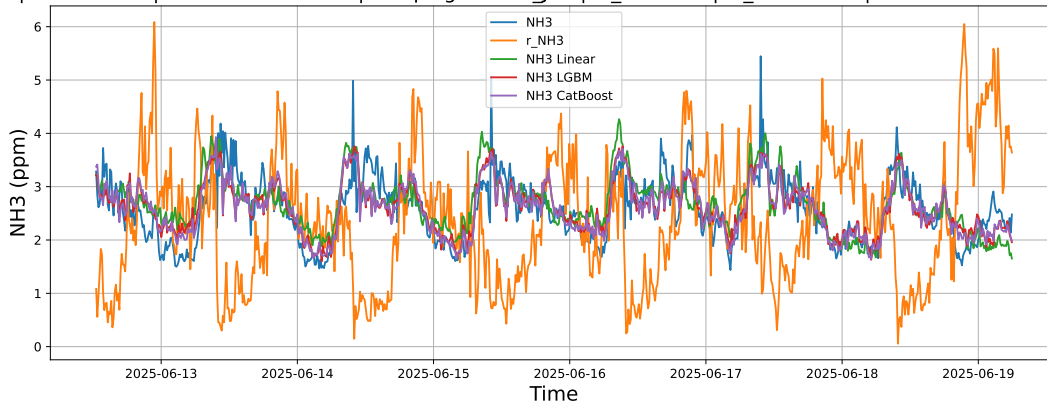
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

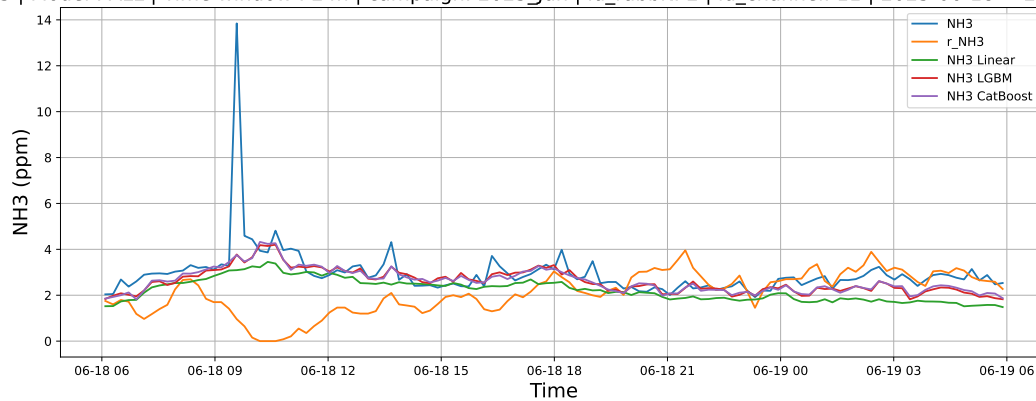
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

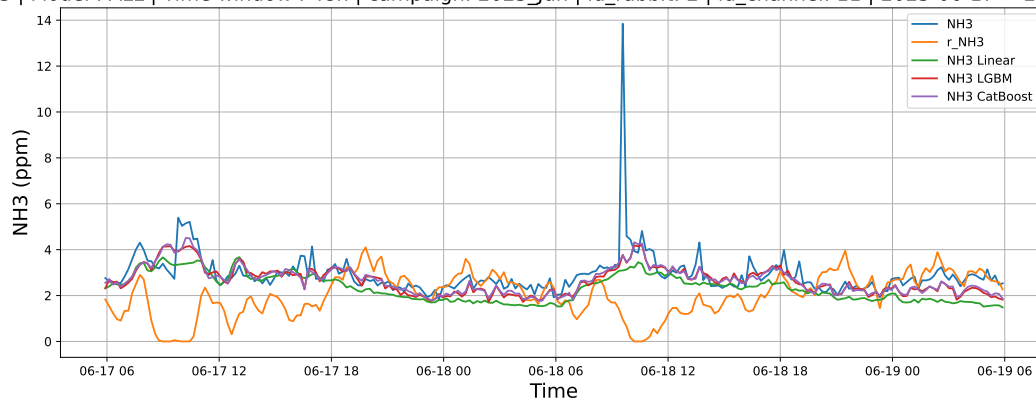
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



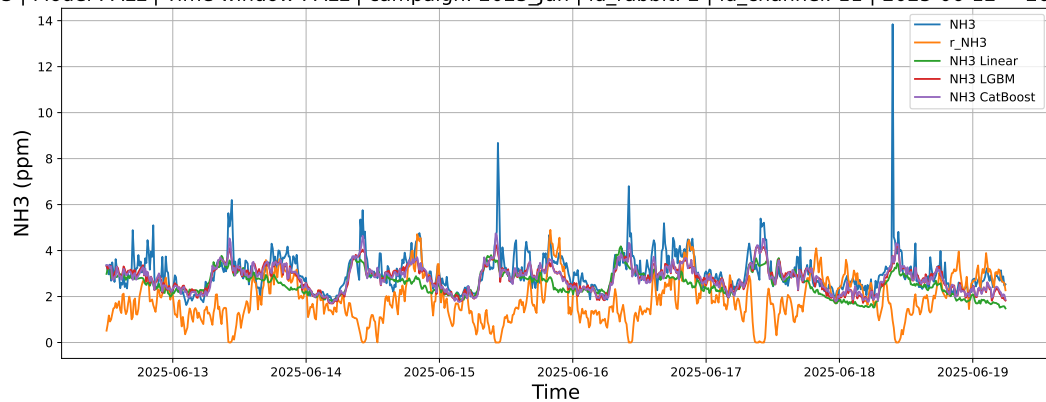
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

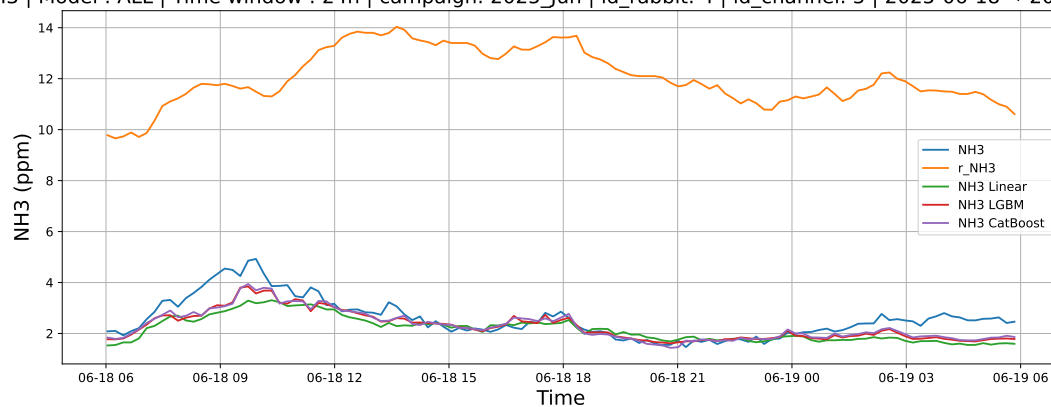
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

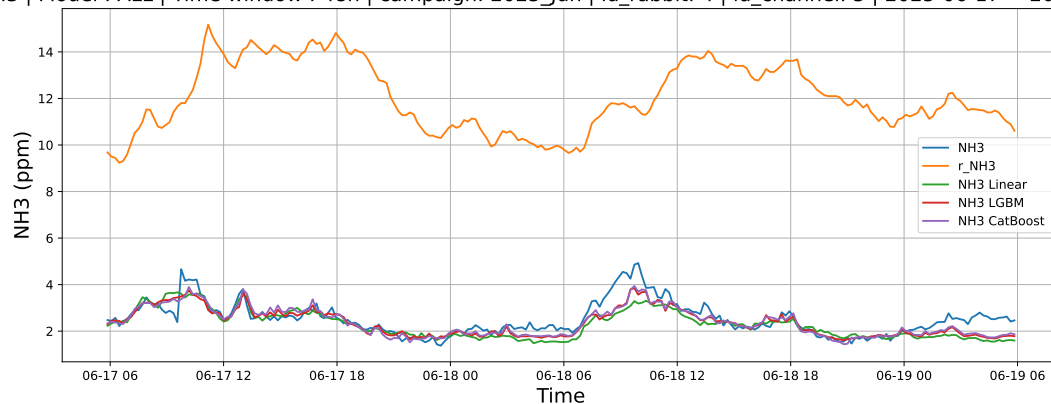
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



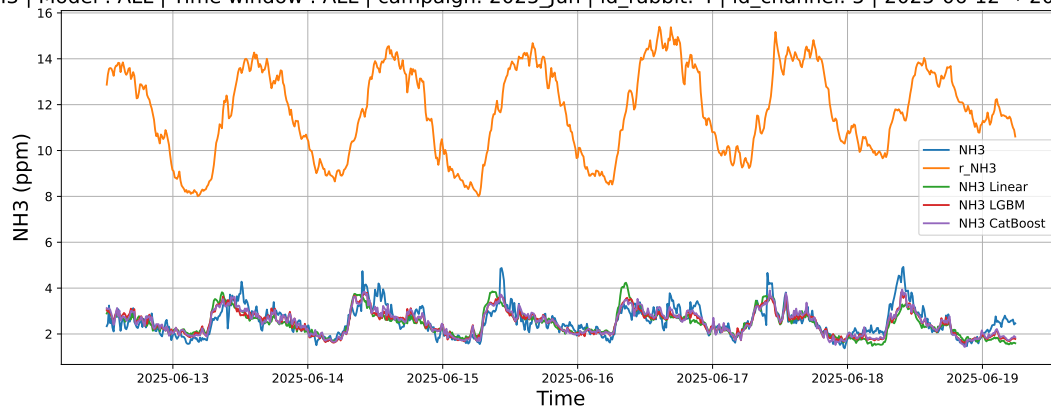
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

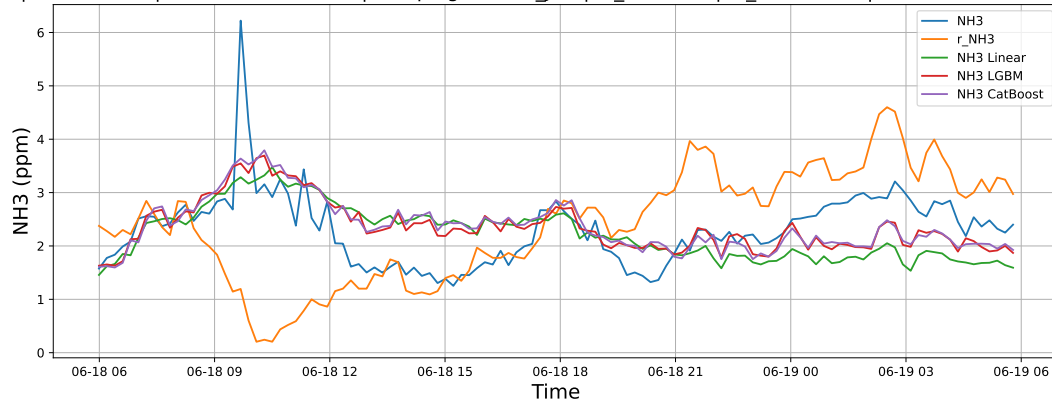
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

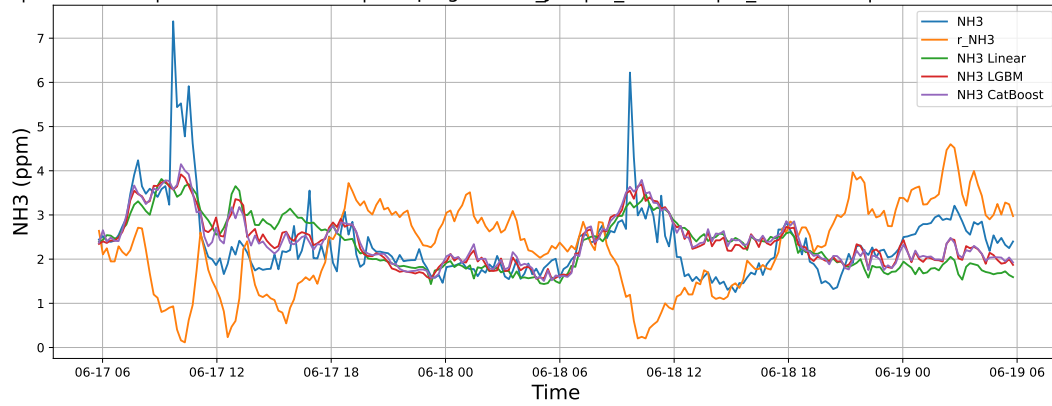
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



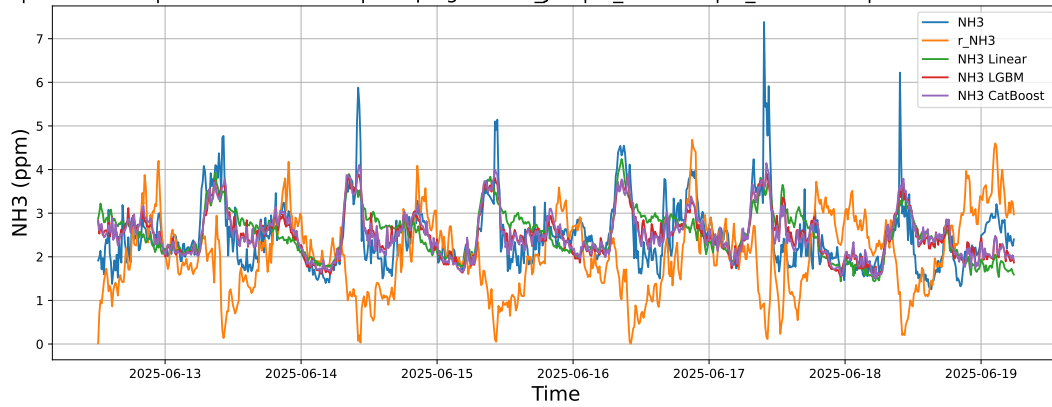
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

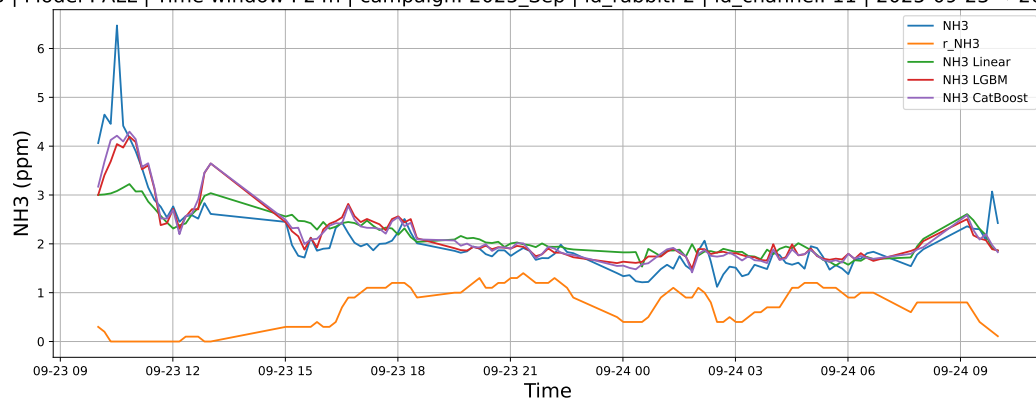
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

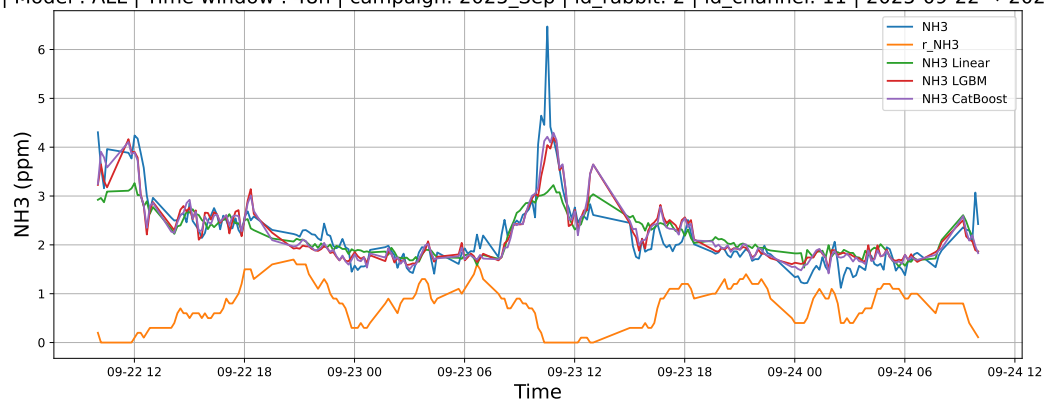
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



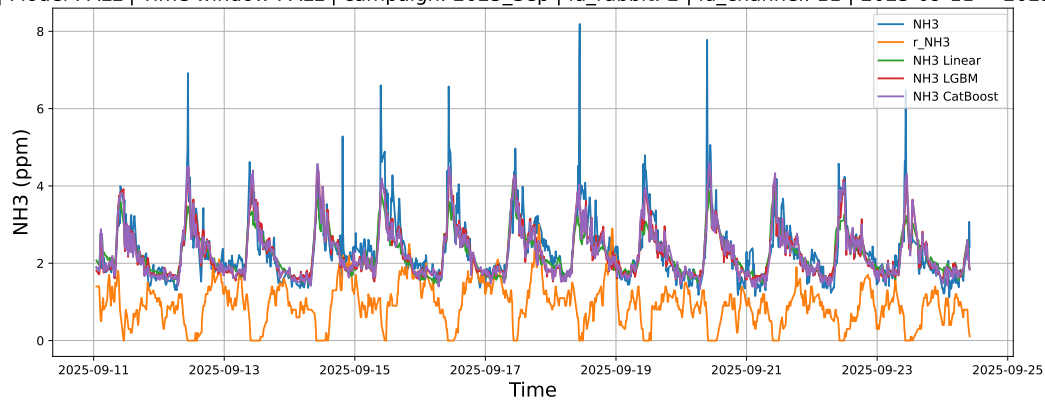
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

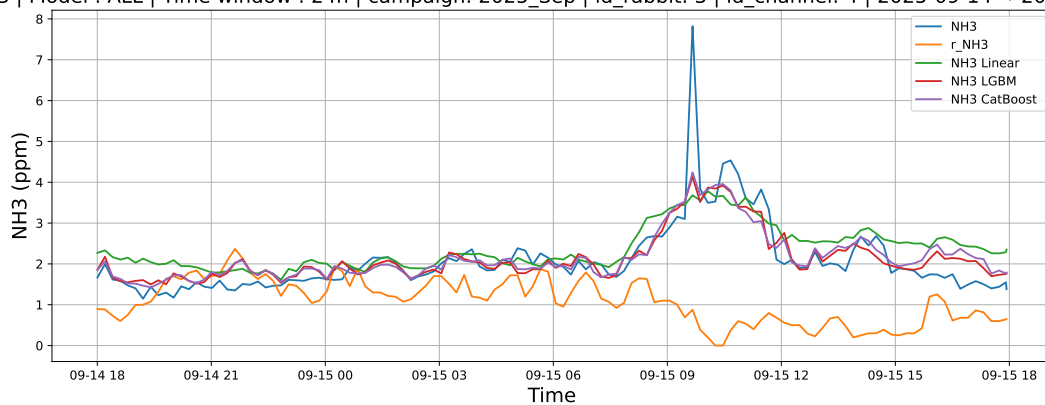
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

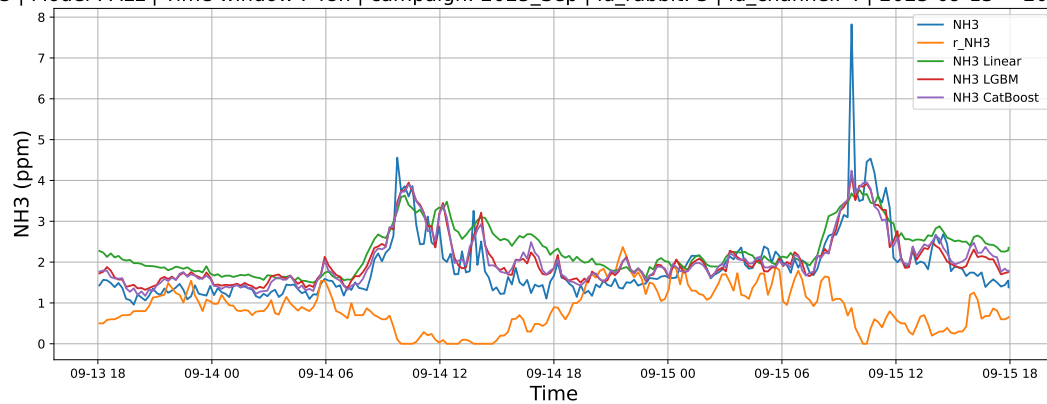
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



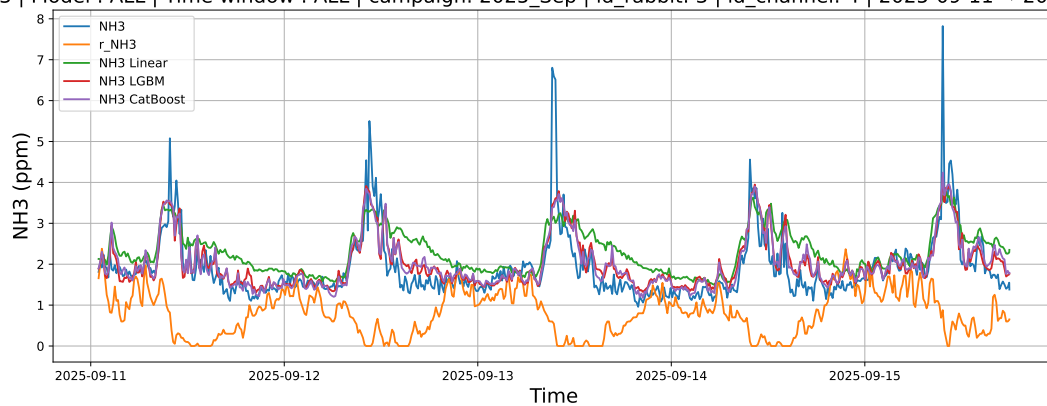
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

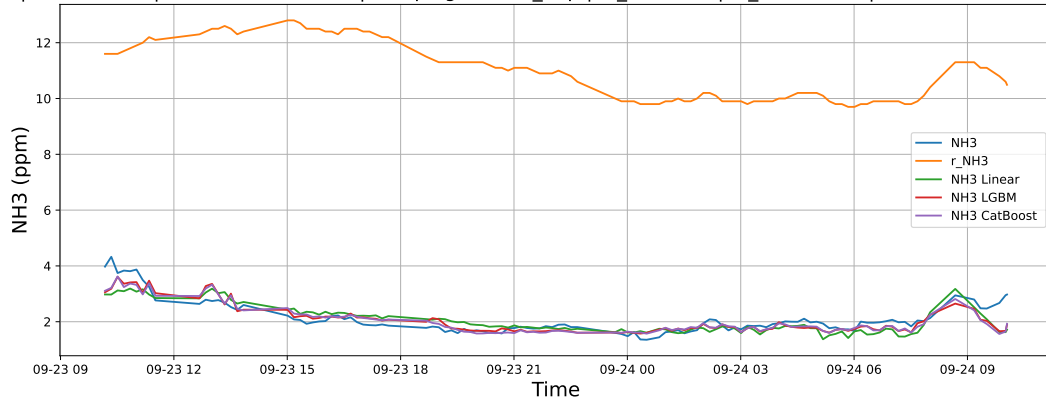
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

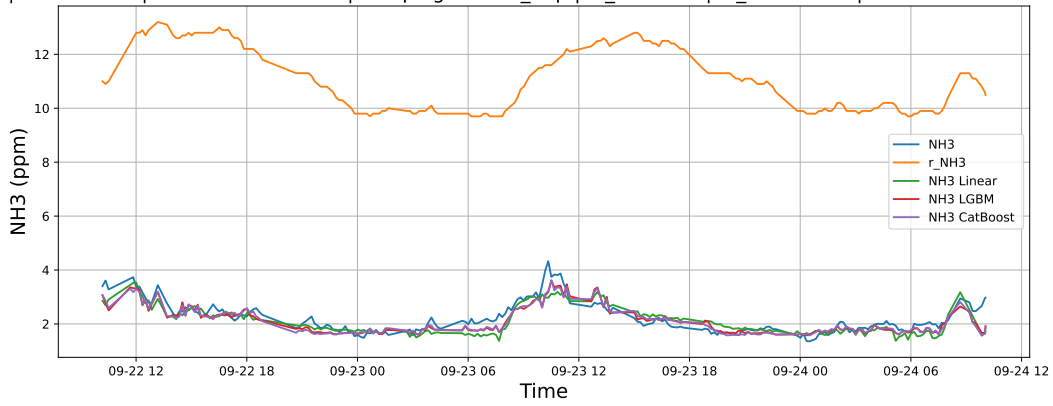
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



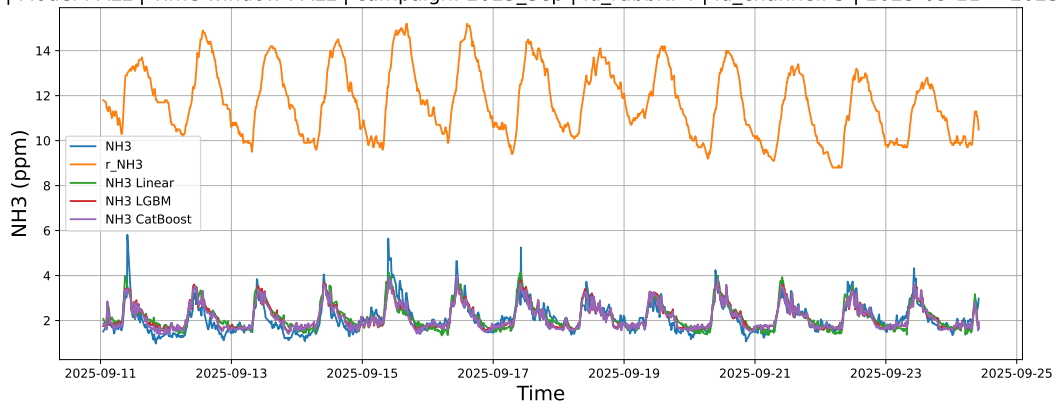
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

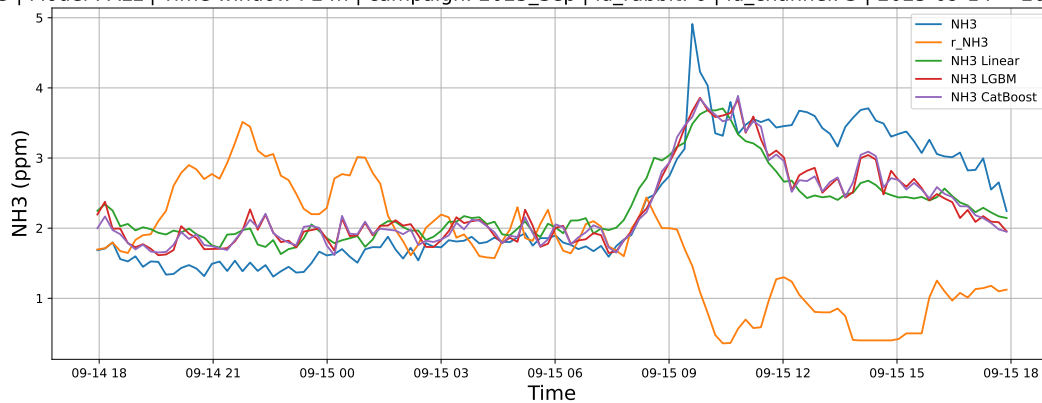
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

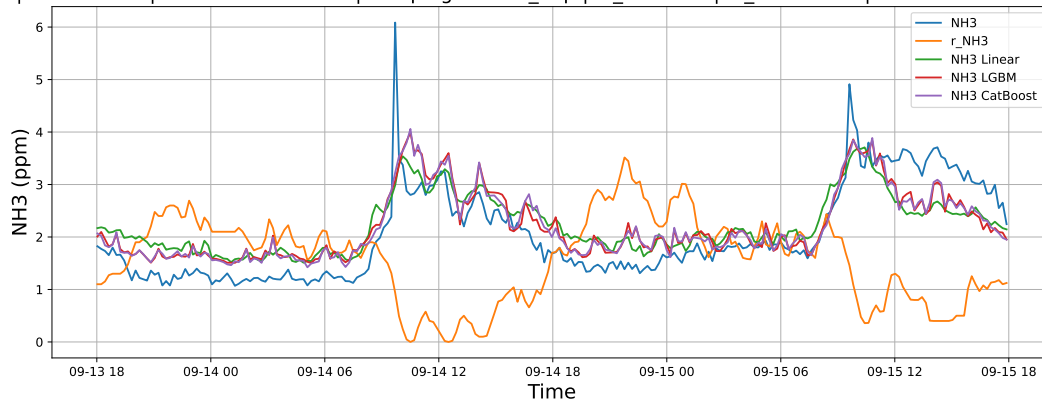
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

